Mini project report on

Disaster Response and Relief Management System

Submitted in partial fulfilment of the requirements for the award of degree of

Bachelor of Technology

in

Computer Science & Engineering

UE23CS351A – DBMS Project

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CERTIFICATE

This is to certify that the mini project entitled

Disaster Response and Relief Management System

is a bonafide work carried out by

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In partial fulfilment for the completion of fifth semester DBMS Project (UE23CS351A) in the Program of Study – Bachelor of Technology in Computer Science and Engineering under rules and regulations of PES University, Bengaluru during the period AUG. 2026 – DEC. 2026. It is certified that all corrections / suggestions indicated for internal assessment have been incorporated in the report. The project has been approved as it satisfies the 5th semester academic requirements in respect of project work.

Signature
Dr. Shilpa S
Assistant Professor

DECLARATION

We hereby declare that the DBMS Project entitled Disaster Response and Relief Management System has been carried out by us under the guidance of Dr. Shilpa S, Assistant Professor and submitted in partial fulfilment of the course requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering of PES University, Bengaluru during the academic semester AUG – DEC 2026.

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ABSTRACT

During natural disasters such as floods and landslides, relief agencies must quickly know who is affected, where they are, what they need, and which shelters, supplies and rescue teams are available. When this information is kept in separate registers or spreadsheets it becomes slow to update, inconsistent and hard to coordinate.

The Disaster Response and Relief Management System is a relational database built in MySQL to solve this problem. It stores information about disasters, affected areas, affected people, relief and medical requests, shelters and shelter allocation, resources and inventory, supply distribution, rescue teams and rescue vehicles in twelve inter-related tables. The design was prepared as an ER model, mapped to a relational schema, and implemented using DDL statements with primary keys, foreign keys, a unique key, CHECK constraints, an index and views.

The report documents the table creation, sample data insertion, ALTER, UPDATE and DELETE operations, and a range of queries – selection, ordering, GROUP BY / HAVING, aggregate functions, inner/left/right joins and nested queries. It also shows a stored function, a stored procedure and two triggers that automate common tasks, such as refreshing the stock-update date and marking a shelter as full when its capacity is reached.

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1. INTRODUCTION

1.1 Overview

Disasters such as floods and landslides affect thousands of people within a short time. Effective response depends on quick access to accurate information: which areas are hit, how many people need help, what supplies are in stock, which shelters have space, and which rescue teams and vehicles are free. A Database Management System (DBMS) gives a central, consistent and secure place to keep and query this information.

This mini project, the Disaster Response and Relief Management System, is a relational database designed and implemented in MySQL. It records disasters and the areas they affect, the people living there and their needs, relief and medical requests, shelters and the allocation of people to them, resources and inventory, supply distribution, and rescue teams with their vehicles.

1.2 Objectives

- To design and develop a Disaster Response and Relief Management System using a DBMS that efficiently stores, organizes and manages disaster-related information.
- To improve coordination and ensure effective management of relief activities during disaster situations.
- To keep data consistent using primary keys, foreign keys and constraints.
- To retrieve useful information (priority areas, shelter capacity, stock levels, team availability) through SQL queries.
- To automate routine checks using stored functions, procedures and triggers.

1.3 Scope

The system covers the data side of relief management: twelve tables, sample data for five disasters in Karnataka (Kodagu, Chikkamagaluru and Sakleshpura landslides; Bengaluru urban and Dakshina Kannada floods), SQL operations on that data, and database programs (function, procedure and triggers).

1.4 Tools Used

Component	Details
Database	MySQL (MySQL command-line client)
Language	SQL (DDL, DML, queries, PL/SQL-style stored programs)
Design	Hand-drawn ER diagram and schema diagram

2. PROBLEM DEFINITION

2.1 Problem Statement

Relief organizations often keep disaster information in separate registers and spreadsheets. This causes duplicate or conflicting records, slow updates and poor visibility of what is needed and where. As a result, supplies may not reach the people with the highest priority, shelters may be over-filled while others stay empty, and rescue teams may be sent without knowing which vehicles are available.

There is a need for a single database that links disasters, affected areas and people with their requests, and links those requests with shelters, stock and distribution, so that decisions can be made quickly and accurately.

2.2 Requirements

- **Disaster records:** name, type, start and end date, severity, status and description.
- **Affected areas and people:** area, location, people affected and priority; each person's age, gender, contact, address, need type and priority.
- **Requests:** relief requests (shelter, food, medical, water) and medical requests with priority, status and the doctor assigned.
- **Shelters:** capacity, current occupancy, type, status and the allocation of people to shelters.
- **Resources:** resource master list, stock quantity and location, and records of supplies distributed.
- **Rescue operations:** rescue teams, team leaders and the vehicles assigned to each team.

2.3 Proposed Solution

A normalized relational database with twelve tables linked by foreign keys. SQL queries answer operational questions and triggers/functions keep important values (such as stock-update date and shelter status) correct automatically.

3. ER MODEL

3.1 ER Diagram

The Entity-Relationship model below shows the entities of the system, their attributes and the relationships between them.

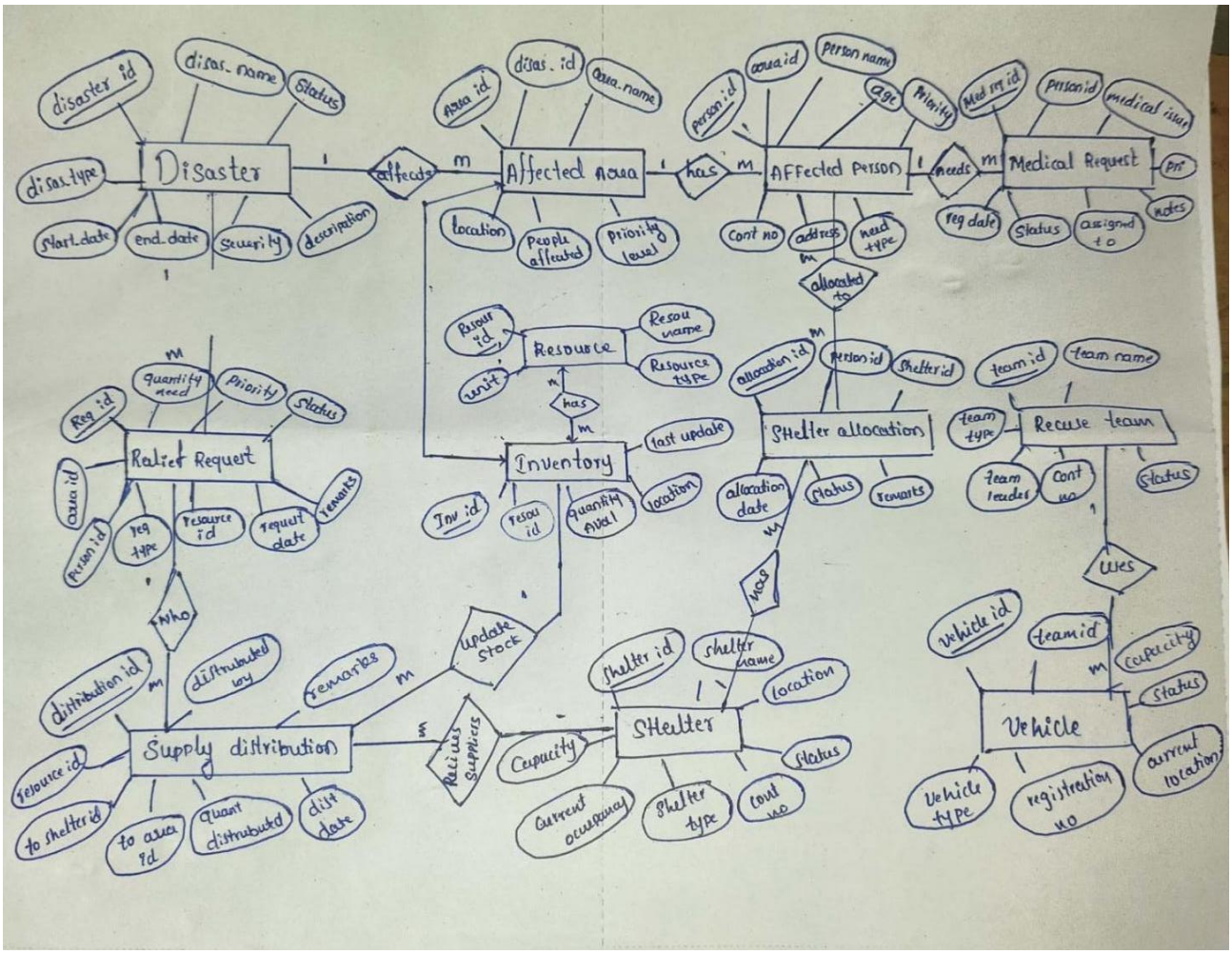


Figure 3.1: ER diagram of the Disaster Response and Relief Management System

3.2 Entities and Attributes

Table 3.1: Entities and their attributes

Entity	Attributes
Disaster	disaster_id, disaster_name, disaster_type, start_date, end_date, severity, status, description
Affected Area	area_id, disaster_id, area_name, location, people_affected, priority_level
Affected Person	person_id, area_id, person_name, age, gender, contact_no, address, need_type, priority
Relief Request	request_id, area_id, person_id, resource_id, request_type, quantity_needed, priority, request_date, status, remarks

Entity	Attributes
Medical Request	medical_request_id, person_id, medical_issue, priority, request_date, status, assigned_to, notes
Shelter	shelter_id, shelter_name, location, capacity, current_occupancy, shelter_type, contact_no, status
Shelter Allocation	allocation_id, person_id, shelter_id, allocation_date, status, remarks
Resource	resource_id, resource_name, resource_type, unit
Inventory	inventory_id, resource_id, quantity_available, last_updated, location
Supply Distribution	distribution_id, resource_id, shelter_id, area_id, quantity_distributed, distribution_date, distributed_by, remarks
Rescue Team	team_id, team_name, team_type, team_leader, contact_no, status
Vehicle	vehicle_id, team_id, vehicle_type, registration_no, capacity, status, current_location

3.3 Relationships

Table 3.2: Relationships between entities

Relationship	Entities	Cardinality
Affects	Disaster – Affected Area	1 : N
Has	Affected Area – Affected Person	1 : N
Needs	Affected Person – Medical Request	1 : N
Raises (requests)	Affected Area / Person / Resource – Relief Request	1 : N
Allocated to	Affected Person – Shelter (via Shelter Allocation)	M : N
Has (stock)	Resource – Inventory	1 : N
Distributes	Resource / Shelter / Area – Supply Distribution	1 : N
Uses	Rescue Team – Vehicle	1 : N

4. ER TO RELATIONAL MAPPING

4.1 Steps of Algorithm for the Chosen Problem

The ER diagram was converted to relations using the standard seven-step ER-to-relational mapping algorithm.

- **Step 1 – Regular entity types.** Every entity became a relation with its simple attributes. The identifier of each entity became the primary key (disaster_id, area_id, person_id, request_id, medical_request_id, shelter_id, allocation_id, resource_id, inventory_id, distribution_id, team_id, vehicle_id).
- **Step 2 – Weak entity types.** None. Every entity has its own key.
- **Step 3 – Binary 1:1 relationships.** None in this design.
- **Step 4 – Binary 1:N relationships.** The primary key of the "1" side was added as a foreign key on the "N" side: disaster_id in Affected Area; area_id in Affected Person; person_id in Medical Request; resource_id in Inventory; team_id in Vehicle; and area_id, person_id, resource_id in Relief Request.
- **Step 5 – Binary M:N relationships.** The Affected Person – Shelter relationship became the relation Shelter Allocation with foreign keys person_id and shelter_id. Supply Distribution links Resource with Shelter and Affected Area through resource_id, shelter_id and area_id.
- **Step 6 – Multivalued attributes.** None.
- **Step 7 – N-ary relationships.** None.

4.2 Complete Diagram of Relational Mapping

The schema diagram below shows each relation with its attributes and the foreign key references between them.

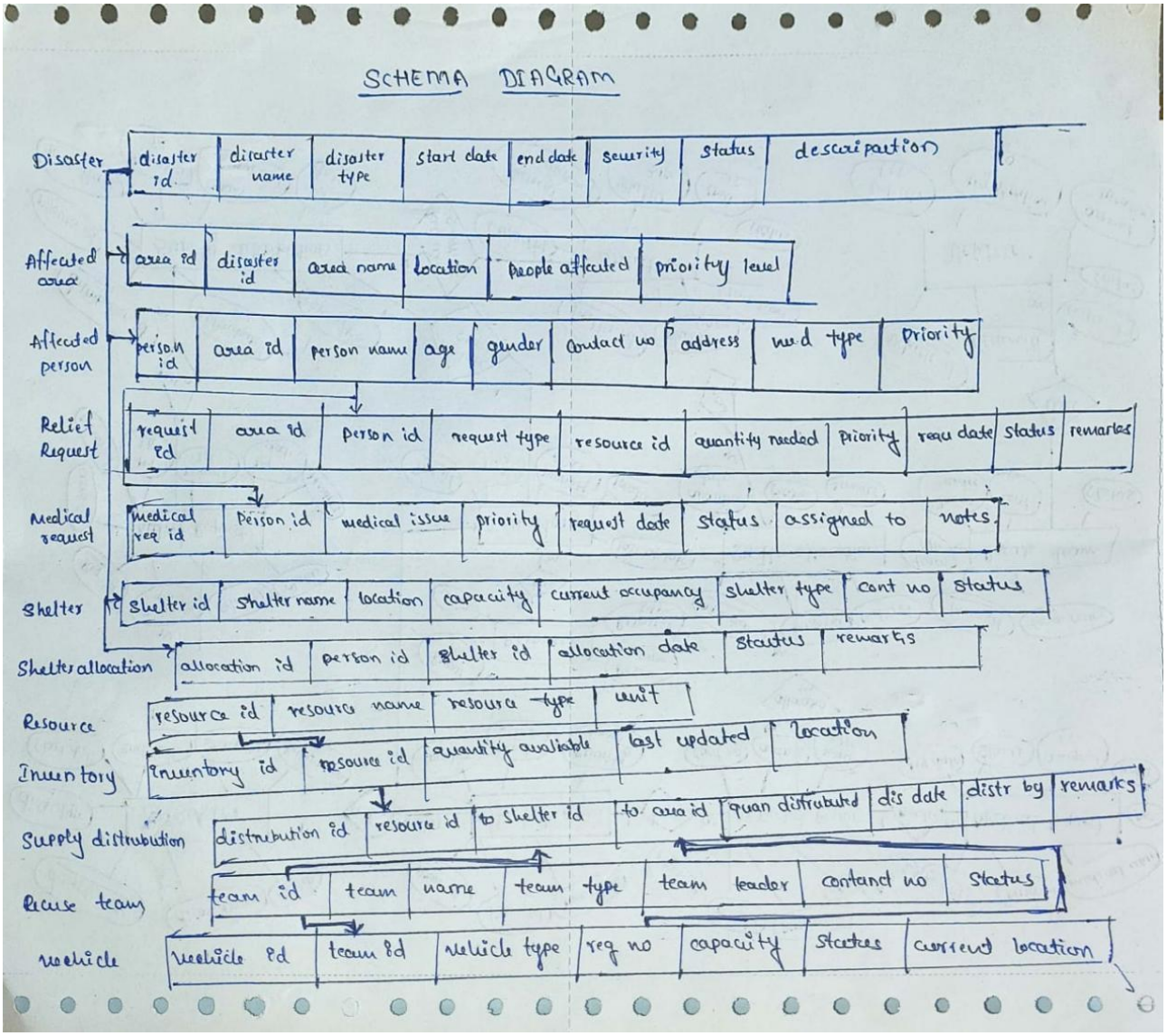


Figure 4.1: Schema diagram (relational mapping)

Table 4.1: Relational schema with primary and foreign keys

Relation	Primary key	Foreign key(s)
disaster	disaster_id	—
affected_area	area_id	disaster_id → disaster
affected_person	person_id	area_id → affected_area
relief_request	request_id	area_id → affected_area; person_id → affected_person; resource_id → resource
medical_request	medical_request_id	person_id → affected_person
shelter	shelter_id	—
shelter_allocation	allocation_id	person_id → affected_person; shelter_id → shelter
resource	resource_id	—
inventory	inventory_id	resource_id → resource
supply_distribution	distribution_id	resource_id → resource; shelter_id → shelter; area_id → affected_area

Relation	Primary key	Foreign key(s)
rescue_team	team_id	—
vehicle	vehicle_id	team_id → rescue_team

5. DDL STATEMENTS

5.1 Table Creation

The database disaster_relief_management contains twelve tables. The screenshot below lists them using SHOW TABLES, followed by the structure of each table (DESC), which shows the column names, data types, NULL rules, keys (PRI = primary key, MUL = foreign key / indexed, UNI = unique) and AUTO_INCREMENT on every primary key.

```
PES2UG24CS463 >show tables;
+---------------------------------------+
| Tables_in_disaster_relief_management |
+---------------------------------------+
| affected_area                         |
| affected_person                       |
| disaster                              |
| inventory                             |
| medical_request                       |
| relief_request                        |
| rescue_team                           |
| resource                              |
| shelter                               |
| shelter_allocation                    |
| supply_distribution                   |
| vehicle                               |
+---------------------------------------+
12 rows in set (0.06 sec)
```

Figure 5.1: Tables created in the database (SHOW TABLES)

```
PES2UG24CS463 >DESC disaster;
+---------------+--------------+------+-----+---------+----------------+
| Field         | Type         | Null | Key | Default | Extra          |
+---------------+--------------+------+-----+---------+----------------+
| disaster_id   | int          | NO   | PRI | NULL    | auto_increment |
| disaster_name | varchar(100) | NO   |     | NULL    |                |
| disaster_type | varchar(100) | YES  |     | NULL    |                |
| start_date    | date         | YES  |     | NULL    |                |
| end_date      | date         | YES  |     | NULL    |                |
| severity      | varchar(50)  | YES  |     | NULL    |                |
| status        | varchar(50)  | YES  |     | NULL    |                |
| description   | text         | YES  |     | NULL    |                |
+---------------+--------------+------+-----+---------+----------------+
8 rows in set (0.15 sec)
```

Figure 5.2: Structure of the disaster table (DESC disaster)

```
PES2UG24CS463 >DESC affected_area;
+------------------+--------------+------+-----+---------+----------------+
| Field            | Type         | Null | Key | Default | Extra          |
+------------------+--------------+------+-----+---------+----------------+
| area_id          | int          | NO   | PRI | NULL    | auto_increment |
| disaster_id      | int          | NO   | MUL | NULL    |                |
| area_name        | varchar(100) | NO   |     | NULL    |                |
| location         | varchar(200) | YES  |     | NULL    |                |
| people_affected  | int          | YES  |     | NULL    |                |
| priority_level   | varchar(50)  | YES  |     | NULL    |                |
+------------------+--------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

Figure 5.3: Structure of the affected_area table (DESC affected_area)

```
PES2UG24CS463 >DESC affected_person;
+-------------+--------------+------+-----+---------+----------------+
| Field       | Type         | Null | Key | Default | Extra          |
+-------------+--------------+------+-----+---------+----------------+
| person_id   | int          | NO   | PRI | NULL    | auto_increment |
| area_id     | int          | NO   | MUL | NULL    |                |
| person_name | varchar(100) | NO   |     | NULL    |                |
| age         | int          | YES  |     | NULL    |                |
| gender      | varchar(20)  | YES  |     | NULL    |                |
| contact_no  | varchar(20)  | YES  |     | NULL    |                |
| address     | varchar(255) | YES  |     | NULL    |                |
| need_type   | varchar(100) | YES  |     | NULL    |                |
| priority    | varchar(50)  | YES  |     | NULL    |                |
+-------------+--------------+------+-----+---------+----------------+
9 rows in set (0.06 sec)
```

Figure 5.4: Structure of the affected_person table (DESC affected_person)

```
PES2UG24CS463 >DESC shelter;
+-------------------+--------------+------+-----+---------+----------------+
| Field             | Type         | Null | Key | Default | Extra          |
+-------------------+--------------+------+-----+---------+----------------+
| shelter_id        | int          | NO   | PRI | NULL    | auto_increment |
| shelter_name      | varchar(100) | NO   |     | NULL    |                |
| location          | varchar(200) | YES  |     | NULL    |                |
| capacity          | int          | YES  |     | NULL    |                |
| current_occupancy | int          | YES  |     | 0       |                |
| shelter_type      | varchar(100) | YES  |     | NULL    |                |
| contact_no        | varchar(20)  | YES  |     | NULL    |                |
| status            | varchar(50)  | YES  |     | NULL    |                |
+-------------------+--------------+------+-----+---------+----------------+
8 rows in set (0.06 sec)
```

Figure 5.5: Structure of the shelter table (DESC shelter)

```
PES2UG24CS463 >DESC shelter_allocation;
+-----------------+-------------+------+-----+---------+----------------+
| Field           | Type        | Null | Key | Default | Extra          |
+-----------------+-------------+------+-----+---------+----------------+
| allocation_id   | int         | NO   | PRI | NULL    | auto_increment |
| person_id       | int         | NO   | MUL | NULL    |                |
| shelter_id      | int         | NO   | MUL | NULL    |                |
| allocation_date | date        | YES  |     | NULL    |                |
| status          | varchar(50) | YES  |     | NULL    |                |
| remarks         | text        | YES  |     | NULL    |                |
+-----------------+-------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

Figure 5.6: Structure of the shelter_allocation table (DESC shelter_allocation)

```
PES2UG24CS463 >DESC resource;
+---------------+--------------+------+-----+---------+----------------+
| Field         | Type         | Null | Key | Default | Extra          |
+---------------+--------------+------+-----+---------+----------------+
| resource_id   | int          | NO   | PRI | NULL    | auto_increment |
| resource_name | varchar(100) | NO   |     | NULL    |                |
| resource_type | varchar(100) | YES  |     | NULL    |                |
| unit          | varchar(50)  | YES  |     | NULL    |                |
+---------------+--------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)
```

Figure 5.7: Structure of the resource table (DESC resource)

```
PES2UG24CS463 >DESC inventory;
+--------------------+--------------+------+-----+---------+----------------+
| Field              | Type         | Null | Key | Default | Extra          |
+--------------------+--------------+------+-----+---------+----------------+
| inventory_id       | int          | NO   | PRI | NULL    | auto_increment |
| resource_id        | int          | NO   | MUL | NULL    |                |
| quantity_available | int          | YES  |     | 0       |                |
| last_updated       | date         | YES  |     | NULL    |                |
| location           | varchar(200) | YES  |     | NULL    |                |
+--------------------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

Figure 5.8: Structure of the inventory table (DESC inventory)

```
PES2UG24CS463 >DESC supply_distribution;
+----------------------+--------------+------+-----+---------+----------------+
| Field                | Type         | Null | Key | Default | Extra          |
+----------------------+--------------+------+-----+---------+----------------+
| distribution_id      | int          | NO   | PRI | NULL    | auto_increment |
| resource_id          | int          | NO   | MUL | NULL    |                |
| shelter_id           | int          | YES  | MUL | NULL    |                |
| area_id              | int          | YES  | MUL | NULL    |                |
| quantity_distributed | int          | NO   |     | NULL    |                |
| distribution_date    | date         | YES  |     | NULL    |                |
| distributed_by       | varchar(100) | YES  |     | NULL    |                |
| remarks              | text         | YES  |     | NULL    |                |
+----------------------+--------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

Figure 5.9: Structure of the supply_distribution table (DESC supply_distribution)

```
PES2UG24CS463 >DESC rescue_team;
+-------------+--------------+------+-----+---------+----------------+
| Field       | Type         | Null | Key | Default | Extra          |
+-------------+--------------+------+-----+---------+----------------+
| team_id     | int          | NO   | PRI | NULL    | auto_increment |
| team_name   | varchar(100) | NO   |     | NULL    |                |
| team_type   | varchar(100) | YES  |     | NULL    |                |
| team_leader | varchar(100) | YES  |     | NULL    |                |
| contact_no  | varchar(20)  | YES  |     | NULL    |                |
| status      | varchar(50)  | YES  |     | NULL    |                |
+-------------+--------------+------+-----+---------+----------------+
6 rows in set (0.01 sec)
```

Figure 5.10: Structure of the rescue_team table (DESC rescue_team)

```
PES2UG24CS463 >DESC vehicle;
+-----------------+--------------+------+-----+---------+----------------+
| Field           | Type         | Null | Key | Default | Extra          |
+-----------------+--------------+------+-----+---------+----------------+
| vehicle_id      | int          | NO   | PRI | NULL    | auto_increment |
| team_id         | int          | NO   | MUL | NULL    |                |
| vehicle_type    | varchar(100) | YES  |     | NULL    |                |
| registration_no | varchar(50)  | YES  | UNI | NULL    |                |
| capacity        | int          | YES  |     | NULL    |                |
| status          | varchar(50)  | YES  |     | NULL    |                |
| current_location | varchar(200) | YES  |     | NULL    |                |
+-----------------+--------------+------+-----+---------+----------------+
7 rows in set (0.00 sec)
```

Figure 5.11: Structure of the vehicle table (DESC vehicle)

```
PES2UG24CS463 >DESC relief_request;
+-----------------+--------------+------+-----+---------+----------------+
| Field           | Type         | Null | Key | Default | Extra          |
+-----------------+--------------+------+-----+---------+----------------+
| request_id      | int          | NO   | PRI | NULL    | auto_increment |
| area_id         | int          | NO   | MUL | NULL    |                |
| person_id       | int          | NO   | MUL | NULL    |                |
| resource_id     | int          | YES  | MUL | NULL    |                |
| request_type    | varchar(100) | YES  |     | NULL    |                |
| quantity_needed | int          | YES  |     | NULL    |                |
| priority        | varchar(50)  | YES  |     | NULL    |                |
| request_date    | date         | YES  |     | NULL    |                |
| status          | varchar(50)  | YES  |     | NULL    |                |
| remarks         | text         | YES  |     | NULL    |                |
+-----------------+--------------+------+-----+---------+----------------+
10 rows in set (0.00 sec)
```

Figure 5.12: Structure of the relief_request table (DESC relief_request)

```
PES2UG24CS463 >DESC medical_request;
+--------------------+--------------+------+-----+---------+----------------+
| Field              | Type         | Null | Key | Default | Extra          |
+--------------------+--------------+------+-----+---------+----------------+
| medical_request_id | int          | NO   | PRI | NULL    | auto_increment |
| person_id          | int          | NO   | MUL | NULL    |                |
| medical_issue      | varchar(255) | YES  |     | NULL    |                |
| priority           | varchar(50)  | YES  |     | NULL    |                |
| request_date       | date         | YES  |     | NULL    |                |
| status             | varchar(50)  | YES  |     | NULL    |                |
| assigned_to        | varchar(100) | YES  |     | NULL    |                |
| notes              | text         | YES  |     | NULL    |                |
+--------------------+--------------+------+-----+---------+----------------+
8 rows in set (0.00 sec)
```

Figure 5.13: Structure of the medical_request table (DESC medical_request)

5.2 ALTER Statements

New columns were added to existing tables using ALTER TABLE, and the new columns were then filled with UPDATE.

Disaster table – add emergency_contact

```
PES2UG24CS457 PES2UG24CS463 >ALTER TABLE disaster
    -> ADD COLUMN emergency_contact VARCHAR(15);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >desc  disaster;
+-------------------+--------------+------+-----+---------+----------------+
| Field             | Type         | Null | Key | Default | Extra          |
+-------------------+--------------+------+-----+---------+----------------+
| disaster_id       | int          | NO   | PRI | NULL    | auto_increment |
| disaster_name     | varchar(100) | NO   |     | NULL    |                |
| disaster_type     | varchar(100) | YES  |     | NULL    |                |
| start_date        | date         | YES  |     | NULL    |                |
| end_date          | date         | YES  |     | NULL    |                |
| severity          | varchar(50)  | YES  |     | NULL    |                |
| status            | varchar(50)  | YES  |     | NULL    |                |
| description       | text         | YES  |     | NULL    |                |
| emergency_contact | varchar(15)  | YES  |     | NULL    |                |
+-------------------+--------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)
```

Figure 5.14: ALTER TABLE disaster ADD COLUMN emergency_contact

```
PES2UG24CS457 PES2UG24CS463 >select * from disaster;
+-------------+-----------------------+--------------+------------+------------+----------+--------+-----------------------------------+-------------------+
| disaster_id | disaster_name         | disaster_type | start_date | end_date   | severity | status | description                       | emergency_contact |
+-------------+-----------------------+--------------+------------+------------+----------+--------+-----------------------------------+-------------------+
|           1 | Kodagu landslide      | Landslide     | 2026-08-20 | 2026-08-27 | high     | Active | landslide in hilly region of kadagu | 108               |
|           2 | chikamangluru landslide | Landslide   | 2026-08-22 | 2026-08-27 | medium   | Active | landslide in  chikamangluru       | 108               |
|           3 | Saklespura landslide  | Landslide     | 2026-08-19 | 2026-08-25 | medium   | closed | landslide in  Sakalespura         | 108               |
|           4 |  Bengluru urban flood | Flood         | 2026-08-24 | 2026-08-27 | high     | Active | Flood in bengaluru urban          | 108               |
|           5 |  Dakshina kannadaflood | Flood        | 2026-08-22 | 2026-08-26 | medium   | closed | Flood in Dakshina Kannada         | 108               |
+-------------+-----------------------+--------------+------------+------------+----------+--------+-----------------------------------+-------------------+
5 rows in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >
```

Figure 5.15: Disaster table after adding the column

Inventory table – add storage_condition

```
PES2UG24CS457 PES2UG24CS463 >ALTER TABLE inventory
    -> ADD COLUMN storage_condition VARCHAR(50);
Query OK, 0 rows affected (0.08 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >desc inventory;
+--------------------+--------------+------+-----+---------+----------------+
| Field              | Type         | Null | Key | Default | Extra          |
+--------------------+--------------+------+-----+---------+----------------+
| inventory_id       | int          | NO   | PRI | NULL    | auto_increment |
| resource_id        | int          | NO   | MUL | NULL    |                |
| quantity_available | int          | YES  |     | 0       |                |
| last_updated       | date         | YES  |     | NULL    |                |
| location           | varchar(200) | YES  |     | NULL    |                |
| storage_condition  | varchar(50)  | YES  |     | NULL    |                |
+--------------------+--------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >
```

Figure 5.16: ALTER TABLE inventory ADD COLUMN storage_condition

```
PES2UG24CS457 PES2UG24CS463 >UPDATE inventory
    -> SET storage_condition = 'Good';
Query OK, 6 rows affected (0.05 sec)
Rows matched: 6  Changed: 6  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >SELECT * FROM inventory;
+--------------+-------------+-------------------+--------------+-------------------------------+-------------------+
| inventory_id | resource_id | quantity_available | last_updated | location                      | storage_condition |
+--------------+-------------+-------------------+--------------+-------------------------------+-------------------+
|            1 |           1 |              5000 | 2026-08-25   | Kodagu Relief Center          | Good              |
|            2 |           2 |             10000 | 2026-08-25   | Kodagu Relief Center          | Good              |
|            3 |           3 |               500 | 2026-08-26   | Chikkamagaluru Relief Center  | Good              |
|            4 |           4 |              2000 | 2026-08-26   | Hassan Relief Center          | Good              |
|            5 |           5 |               800 | 2026-08-27   | Bengaluru Relief Center       | Good              |
|            6 |           6 |               300 | 2026-08-27   | Dakshina Kannada Relief Center | Good              |
+--------------+-------------+-------------------+--------------+-------------------------------+-------------------+
6 rows in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >
```

Figure 5.17: UPDATE of storage_condition and result

Affected_person table – add emergency_contact

```
PES2UG24CS457 PES2UG24CS463 >ALTER TABLE affected_person
    -> ADD COLUMN emergency_contact VARCHAR(15);
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >DESC attected_person;
ERROR 1146 (42S02): Table 'disaster_relief_management.attected_person' doesn't exist
PES2UG24CS457 PES2UG24CS463 >DESC affected_person;
+-------------------+--------------+------+-----+---------+----------------+
| Field             | Type         | Null | Key | Default | Extra          |
+-------------------+--------------+------+-----+---------+----------------+
| person_id         | int          | NO   | PRI | NULL    | auto_increment |
| area_id           | int          | NO   | MUL | NULL    |                |
| person_name       | varchar(100) | NO   |     | NULL    |                |
| age               | int          | YES  |     | NULL    |                |
| gender            | varchar(20)  | YES  |     | NULL    |                |
| contact_no        | varchar(20)  | YES  |     | NULL    |                |
| address           | varchar(255) | YES  |     | NULL    |                |
| need_type         | varchar(100) | YES  |     | NULL    |                |
| priority          | varchar(50)  | YES  |     | NULL    |                |
| emergency_contact | varchar(15)  | YES  |     | NULL    |                |
+-------------------+--------------+------+-----+---------+----------------+
10 rows in set (0.00 sec)
```

Figure 5.18: ALTER TABLE affected_person ADD COLUMN emergency_contact

```
PES2UG24CS457 PES2UG24CS463 >UPDATE affected_person
    -> SET emergency_contact = contact_no;
Query OK, 10 rows affected (0.05 sec)
Rows matched: 10  Changed: 10  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >select * from affected_person;
+-----------+---------+-----------------+------+--------+------------+---------------------------+-----------+----------+-------------------+
| person_id | area_id | person_name     | age  | gender | contact_no | address                   | need_type | priority | emergency_contact |
+-----------+---------+-----------------+------+--------+------------+---------------------------+-----------+----------+-------------------+
|         1 |       1 | Ramesh Gowda    |   45 | Male   | 9876543201 | Madikeri, Kodagu          | Shelter   | High     | 9876543201        |
|         2 |       1 | Lakshmi Devi    |   38 | Female | 9876543202 | Madikeri, Kodagu          | Medical   | High     | 9876543202        |
|         3 |       2 | Manjunath Kumar |   52 | Male   | 9876543203 | Somwarpet, Kodagu         | Food      | Medium   | 9876543203        |
|         4 |       3 | Suresh Bhat     |   61 | Male   | 9876543204 | Chikkamagaluru            | Medical   | High     | 9876543204        |
|         5 |       3 | Anitha Rao      |   29 | Female | 9876543205 | Chikkamagaluru            | Food      | Medium   | 9876543205        |
|         6 |       4 | Kavya Shetty    |   34 | Female | 9876543206 | Koppa, Chikkamagaluru     | Shelter   | High     | 9876543206        |
|         7 |       5 | Prakash Hegde   |   48 | Male   | 9876543207 | Sakleshpur, Hassan        | Food      | High     | 9876543207        |
|         8 |       7 | Priya Sharma    |   27 | Female | 9876543208 | Whitefield, Bengaluru     | Medical   | High     | 9876543208        |
|         9 |       8 | Arjun Kumar     |   41 | Male   | 9876543209 | Yelahanka, Bengaluru      | Shelter   | Medium   | 9876543209        |
|        10 |       9 | Mohammed Irfan  |   56 | Male   | 9876543210 | Mangaluru, Dakshina Kannada | Food    | High     | 9876543210        |
+-----------+---------+-----------------+------+--------+------------+---------------------------+-----------+----------+-------------------+
10 rows in set (0.00 sec)
```

Figure 5.19: emergency_contact filled from contact_no

Shelter table – add last_inspected

```
PES2UG24CS457 PES2UG24CS463 >ALTER TABLE shelter
    -> ADD COLUMN last_inspected DATE;
Query OK, 0 rows affected (0.10 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >desc shelter;
+--------------------+--------------+------+-----+---------+----------------+
| Field              | Type         | Null | Key | Default | Extra          |
+--------------------+--------------+------+-----+---------+----------------+
| shelter_id         | int          | NO   | PRI | NULL    | auto_increment |
| shelter_name       | varchar(100) | NO   |     | NULL    |                |
| location           | varchar(200) | YES  |     | NULL    |                |
| capacity           | int          | YES  |     | NULL    |                |
| current_occupancy  | int          | YES  |     | 0       |                |
| shelter_type       | varchar(100) | YES  |     | NULL    |                |
| contact_no         | varchar(20)  | YES  |     | NULL    |                |
| status             | varchar(50)  | YES  |     | NULL    |                |
| last_inspected     | date         | YES  |     | NULL    |                |
+--------------------+--------------+------+-----+---------+----------------+
9 rows in set (0.00 sec)
```

Figure 5.20: ALTER TABLE shelter ADD COLUMN last_inspected

5.3 Index

An index on the priority column of affected_person speeds up searches for high-priority people.

```
PES2UG24CS457 PES2UG24CS463 >CREATE INDEX idx_affected_person_priority
    -> ON affected_person(priority);
Query OK, 0 rows affected (0.17 sec)
Records: 0  Duplicates: 0  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >SHOW INDEX FROM affected_person;
+-----------------+------------+--------------------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table           | Non_unique | Key_name                       | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+-----------------+------------+--------------------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| affected_person |          0 | PRIMARY                        |            1 | person_id   | A         |          10 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| affected_person |          1 | area_id                        |            1 | area_id     | A         |           8 |     NULL |   NULL |      | BTREE      |         |               | YES     | NULL       |
| affected_person |          1 | idx_affected_person_priority   |            1 | priority    | A         |           2 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+-----------------+------------+--------------------------------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
3 rows in set (0.06 sec)

PES2UG24CS457 PES2UG24CS463 >
```

Figure 5.21: CREATE INDEX idx_affected_person_priority and SHOW INDEX

5.4 Constraints

CHECK constraints keep data valid: vehicle capacity must be greater than 0, and inventory quantity cannot be negative.

```
3 rows in set (0.06 sec)

PES2UG24CS457 PES2UG24CS463 >ALTER TABLE vehicle
    -> ADD CONSTRAINT chk_vehicle_capacity
    -> CHECK (capacity > 0);
Query OK, 5 rows affected (0.10 sec)
Records: 5  Duplicates: 0  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >SHOW CREATE TABLE vehicle;
+---------+-------------------
```

```
PES2UG24CS457 PES2UG24CS463 >CREATE VIEW high_priority_persons AS
    -> SELECT person_id, person_name, need_type, priority, emergency_contact
    -> FROM affected_person
    -> WHERE priority = 'High';
Query OK, 0 rows affected (0.02 sec)

PES2UG24CS457 PES2UG24CS463 >SELECT * FROM high_priority_persons;
+-----------+---------------+-----------+----------+-------------------+
| person_id | person_name   | need_type | priority | emergency_contact |
+-----------+---------------+-----------+----------+-------------------+
|         1 | Ramesh Gowda  | Shelter   | High     | 9876543201        |
|         2 | Lakshmi Devi  | Medical   | High     | 9876543202        |
|         4 | Suresh Bhat   | Medical   | High     | 9876543204        |
|         6 | Kavya Shetty  | Shelter   | High     | 9876543206        |
|         7 | Prakash Hegde | Food      | High     | 9876543207        |
|         8 | Priya Sharma  | Medical   | High     | 9876543208        |
+-----------+---------------+-----------+----------+-------------------+
6 rows in set (0.05 sec)

PES2UG24CS457 PES2UG24CS463 >
```

Figure 5.24: View high_priority_persons (affected people with High priority)

```
PES2UG24CS457 PES2UG24CS463 >CREATE VIEW active_shelters AS
    -> SELECT shelter_id, shelter_name, location, capacity, current_occupancy, status
    -> FROM shelter
    -> WHERE status = 'Active';
Query OK, 0 rows affected (0.01 sec)

PES2UG24CS457 PES2UG24CS463 >SELECT * FROM active_shelters;
+------------+-------------------------------+----------------+----------+-------------------+--------+
| shelter_id | shelter_name                  | location       | capacity | current_occupancy | status |
+------------+-------------------------------+----------------+----------+-------------------+--------+
|          1 | Kodagu Government School      | Madikeri       |      500 |               350 | Active |
|          2 | Somwarpet Community Hall      | Somwarpet      |      300 |               180 | Active |
|          3 | Chikkamagaluru Relief Camp    | Chikkamagaluru |      600 |               420 | Active |
|          4 | Sakleshpur Government College | Sakleshpur     |      400 |               250 | Active |
|          5 | Bengaluru Relief Center       | Whitefield     |      800 |               650 | Active |
|          6 | Mangaluru Community Center    | Mangaluru      |      700 |               300 | Active |
+------------+-------------------------------+----------------+----------+-------------------+--------+
6 rows in set (0.00 sec)
```

Figure 5.25: View active_shelters (shelters with Active status)

6. DML STATEMENTS

Sample data was inserted into every table using INSERT statements. Each screenshot shows the statement and the table content verified with SELECT *.

6.1 disaster table

```
PES2UG24CS457 PES2UG24CS463 >clear;
PES2UG24CS457 PES2UG24CS463 >insert into disaster(disaster_id,disaster_name,disaster_type,start_date,end_date,severity,status,description)values(1,'Kodagu landslide','Landslide','2026-08-20','2026-08-27','hig
h','Active','landslide in hilly region of kadagu');
Query OK, 1 row affected (0.06 sec)

PES2UG24CS457 PES2UG24CS463 >select *from disaster;
+-------------+-----------------+--------------+------------+------------+----------+--------+-----------------------------------+
| disaster_id | disaster_name    | disaster_type | start_date | end_date   | severity | status | description                       |
+-------------+-----------------+--------------+------------+------------+----------+--------+-----------------------------------+
|           1 | Kodagu landslide | Landslide     | 2026-08-20 | 2026-08-27 | high     | Active | landslide in hilly region of kadagu |
+-------------+-----------------+--------------+------------+------------+----------+--------+-----------------------------------+
1 row in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >insert into disaster(disaster_id,disaster_name,disaster_type,start_date,end_date,severity,status,description)values(2,'chikamangluru landslide','Landslide','2026-08-22','2026-08-2
7','medium','Active','landslide in  chikamangluru');
Query OK, 1 row affected (0.05 sec)

PES2UG24CS457 PES2UG24CS463 >insert into disaster(disaster_id,disaster_name,disaster_type,start_date,end_date,severity,status,description)values(3,'Saklespura landslide','Landslide','2026-08-19','2026-08-25',
'medium','closed','landslide in  Sakalespura');
Query OK, 1 row affected (0.05 sec)

PES2UG24CS457 PES2UG24CS463 >insert into disaster(disaster_id,disaster_name,disaster_type,start_date,end_date,severity,status,description)values(4,' Bengluru urban flood','Flood','2026-08-24','Null','high','A
ctive','Flood in bengaluru urban');
ERROR 1292 (22007): Incorrect date value: 'Null' for column 'end_date' at row 1
PES2UG24CS457 PES2UG24CS463 >insert into disaster(disaster_id,disaster_name,disaster_type,start_date,end_date,severity,status,description)values(4,' Bengluru urban flood','Flood','2026-08-24','2026-08-27','hi
gh','Active','Flood in bengaluru urban');
Query OK, 1 row affected (0.05 sec)

PES2UG24CS457 PES2UG24CS463 >insert into disaster(disaster_id,disaster_name,disaster_type,start_date,end_date,severity,status,description)values(5,' Dakshina kannadaflood','Flood','2026-08-22','2026-08-26','m
edium','closed','Flood in Dakshina Kannada');
Query OK, 1 row affected (0.05 sec)
```

Figure 6.1: INSERT statement for disaster

```
Query OK, 1 row affected (0.05 sec)

PES2UG24CS457 PES2UG24CS463 >select * from disaster;
+-------------+-----------------------+--------------+------------+------------+----------+--------+-----------------------------------+
| disaster_id | disaster_name          | disaster_type | start_date | end_date   | severity | status | description                       |
+-------------+-----------------------+--------------+------------+------------+----------+--------+-----------------------------------+
|           1 | Kodagu landslide       | Landslide     | 2026-08-20 | 2026-08-27 | high     | Active | landslide in hilly region of kadagu |
|           2 | chikamangluru landslide | Landslide     | 2026-08-22 | 2026-08-27 | medium   | Active | landslide in  chikamangluru       |
|           3 | Saklespura landslide   | Landslide     | 2026-08-19 | 2026-08-25 | medium   | closed | landslide in  Sakalespura         |
|           4 |  Bengluru urban flood  | Flood         | 2026-08-24 | 2026-08-27 | high     | Active | Flood in bengaluru urban          |
|           5 |  Dakshina kannadaflood | Flood         | 2026-08-22 | 2026-08-26 | medium   | closed | Flood in Dakshina Kannada         |
+-------------+-----------------------+--------------+------------+------------+----------+--------+-----------------------------------+
5 rows in set (0.00 sec)
```

Figure 6.2: Contents of disaster after insertion

6.2 affected_area table

```
PES2UG24CS457 PES2UG24CS463 >INSERT INTO affected_area
    -> (area_id, disaster_id, area_name, location, people_affected, priority_level)
    -> VALUES
    -> (1, 1, 'Madikeri', 'Kodagu', 900, 'High'),
    -> (2, 1, 'Somwarpet', 'Kodagu', 600, 'Medium'),
    -> (3, 2, 'Chikkamagaluru', 'Chikkamagaluru', 1200, 'High'),
    -> (4, 2, 'Koppa', 'Chikkamagaluru', 700, 'Medium'),
    -> (5, 3, 'Sakleshpur', 'Hassan', 500, 'High'),
    -> (6, 3, 'Belur', 'Hassan', 800, 'Medium'),
    -> (7, 4, 'Whitefield', 'Bengaluru', 1200, 'High'),
    -> (8, 4, 'Yelahanka', 'Bengaluru', 750, 'High'),
    -> (9, 5, 'Mangaluru', 'Dakshina Kannada', 1500, 'High'),
    -> (10, 5, 'Udupi Road', 'Dakshina Kannada', 650, 'Medium');
Query OK, 10 rows affected (0.05 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

Figure 6.3: INSERT statement for affected_area

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM affected_area;
+---------+-------------+----------------+-----------------+-----------------+----------------+
| area_id | disaster_id | area_name      | location        | people_affected | priority_level |
+---------+-------------+----------------+-----------------+-----------------+----------------+
|       1 |           1 | Madikeri       | Kodagu          |             900 | High           |
|       2 |           1 | Somwarpet      | Kodagu          |             600 | Medium         |
|       3 |           2 | Chikkamagaluru | Chikkamagaluru  |            1200 | High           |
|       4 |           2 | Koppa          | Chikkamagaluru  |             700 | Medium         |
|       5 |           3 | Sakleshpur     | Hassan          |             500 | High           |
|       6 |           3 | Belur          | Hassan          |             800 | Medium         |
|       7 |           4 | Whitefield     | Bengaluru       |            1200 | High           |
|       8 |           4 | Yelahanka      | Bengaluru       |             750 | High           |
|       9 |           5 | Mangaluru      | Dakshina Kannada|            1500 | High           |
|      10 |           5 | Udupi Road     | Dakshina Kannada|             650 | Medium         |
+---------+-------------+----------------+-----------------+-----------------+----------------+
10 rows in set (0.00 sec)
```

Figure 6.4: Contents of affected_area after insertion

6.3 affected_person table

```
PES2UG24CS457 PES2UG24CS463 >INSERT INTO affected_person
    -> (person_id, area_id, person_name, age, gender, contact_no, address, need_type, priority)
    -> VALUES
    -> (1, 1, 'Ramesh Gowda', 45, 'Male', '9876543201', 'Madikeri, Kodagu', 'Shelter', 'High'),
    -> (2, 1, 'Lakshmi Devi', 38, 'Female', '9876543202', 'Madikeri, Kodagu', 'Medical', 'High'),
    -> (3, 2, 'Manjunath Kumar', 52, 'Male', '9876543203', 'Somwarpet, Kodagu', 'Food', 'Medium'),
    -> (4, 3, 'Suresh Bhat', 61, 'Male', '9876543204', 'Chikkamagaluru', 'Medical', 'High'),
    -> (5, 3, 'Anitha Rao', 29, 'Female', '9876543205', 'Chikkamagaluru', 'Food', 'Medium'),
    -> (6, 4, 'Kavya Shetty', 34, 'Female', '9876543206', 'Koppa, Chikkamagaluru', 'Shelter', 'High'),
    -> (7, 5, 'Prakash Hegde', 48, 'Male', '9876543207', 'Sakleshpur, Hassan', 'Food', 'High'),
    -> (8, 7, 'Priya Sharma', 27, 'Female', '9876543208', 'Whitefield, Bengaluru', 'Medical', 'High'),
    -> (9, 8, 'Arjun Kumar', 41, 'Male', '9876543209', 'Yelahanka, Bengaluru', 'Shelter', 'Medium'),
    -> (10, 9, 'Mohammed Irfan', 56, 'Male', '9876543210', 'Mangaluru, Dakshina Kannada', 'Food', 'High');
Query OK, 10 rows affected (0.06 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

Figure 6.5: INSERT statement for affected_person

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM affected_person;
+-----------+---------+-----------------+------+--------+------------+---------------------------+-----------+----------+
| person_id | area_id | person_name     | age  | gender | contact_no | address                   | need_type | priority |
+-----------+---------+-----------------+------+--------+------------+---------------------------+-----------+----------+
|         1 |       1 | Ramesh Gowda    |   45 | Male   | 9876543201 | Madikeri, Kodagu          | Shelter   | High     |
|         2 |       1 | Lakshmi Devi    |   38 | Female | 9876543202 | Madikeri, Kodagu          | Medical   | High     |
|         3 |       2 | Manjunath Kumar |   52 | Male   | 9876543203 | Somwarpet, Kodagu         | Food      | Medium   |
|         4 |       3 | Suresh Bhat     |   61 | Male   | 9876543204 | Chikkamagaluru            | Medical   | High     |
|         5 |       3 | Anitha Rao      |   29 | Female | 9876543205 | Chikkamagaluru            | Food      | Medium   |
|         6 |       4 | Kavya Shetty    |   34 | Female | 9876543206 | Koppa, Chikkamagaluru     | Shelter   | High     |
|         7 |       5 | Prakash Hegde   |   48 | Male   | 9876543207 | Sakleshpur, Hassan        | Food      | High     |
|         8 |       7 | Priya Sharma    |   27 | Female | 9876543208 | Whitefield, Bengaluru     | Medical   | High     |
|         9 |       8 | Arjun Kumar     |   41 | Male   | 9876543209 | Yelahanka, Bengaluru      | Shelter   | Medium   |
|        10 |       9 | Mohammed Irfan  |   56 | Male   | 9876543210 | Mangaluru, Dakshina Kannada | Food    | High     |
+-----------+---------+-----------------+------+--------+------------+---------------------------+-----------+----------+
10 rows in set (0.00 sec)
```

Figure 6.6: Contents of affected_person after insertion

6.4 resource table

```
PES2UG24CS457 PES2UG24CS463 >INSERT INTO resource
    -> (resource_id, resource_name, resource_type, unit)
    -> VALUES
    -> (1, 'Rice', 'Food', 'Kg'),
    -> (2, 'Drinking Water', 'Water', 'Litres'),
    -> (3, 'Medical Kit', 'Medical', 'Kits'),
    -> (4, 'Blankets', 'Shelter', 'Pieces'),
    -> (5, 'First Aid Kit', 'Medical', 'Kits'),
    -> (6, 'Tents', 'Shelter', 'Pieces');
Query OK, 6 rows affected (0.06 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

Figure 6.7: INSERT statement for resource

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM resource;
+-------------+----------------+---------------+---------+
| resource_id | resource_name  | resource_type | unit    |
+-------------+----------------+---------------+---------+
|           1 | Rice           | Food          | Kg      |
|           2 | Drinking Water | Water         | Litres  |
|           3 | Medical Kit    | Medical       | Kits    |
|           4 | Blankets       | Shelter       | Pieces  |
|           5 | First Aid Kit  | Medical       | Kits    |
|           6 | Tents          | Shelter       | Pieces  |
+-------------+----------------+---------------+---------+
6 rows in set (0.00 sec)
```

Figure 6.8: Contents of resource after insertion

6.5 inventory table

```
PES2UG24CS457 PES2UG24CS463 >INSERT INTO inventory
    -> (inventory_id, resource_id, quantity_available, last_updated, location)
    -> VALUES
    -> (1, 1, 5000, '2026-08-25', 'Kodagu Relief Center'),
    -> (2, 2, 10000, '2026-08-25', 'Kodagu Relief Center'),
    -> (3, 3, 500, '2026-08-26', 'Chikkamagaluru Relief Center'),
    -> (4, 4, 2000, '2026-08-26', 'Hassan Relief Center'),
    -> (5, 5, 800, '2026-08-27', 'Bengaluru Relief Center'),
    -> (6, 6, 300, '2026-08-27', 'Dakshina Kannada Relief Center');
Query OK, 6 rows affected (0.06 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

Figure 6.9: INSERT statement for inventory

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM inventory;
+--------------+-------------+--------------------+--------------+--------------------------------+
| inventory_id | resource_id | quantity_available | last_updated | location                       |
+--------------+-------------+--------------------+--------------+--------------------------------+
|            1 |           1 |               5000 | 2026-08-25   | Kodagu Relief Center           |
|            2 |           2 |              10000 | 2026-08-25   | Kodagu Relief Center           |
|            3 |           3 |                500 | 2026-08-26   | Chikkamagaluru Relief Center   |
|            4 |           4 |               2000 | 2026-08-26   | Hassan Relief Center           |
|            5 |           5 |                800 | 2026-08-27   | Bengaluru Relief Center        |
|            6 |           6 |                300 | 2026-08-27   | Dakshina Kannada Relief Center |
+--------------+-------------+--------------------+--------------+--------------------------------+
6 rows in set (0.00 sec)
```

Figure 6.10: Contents of inventory after insertion

6.6 shelter table

```
PES2UG24CS457 PES2UG24CS463 >INSERT INTO shelter
    -> (shelter_id, shelter_name, location, capacity, current_occupancy, shelter_type, contact_no, status)
    -> VALUES
    -> (1, 'Kodagu Government School', 'Madikeri', 500, 320, 'Temporary', '9000000001', 'Active'),
    -> (2, 'Somwarpet Community Hall', 'Somwarpet', 300, 180, 'Temporary', '9000000002', 'Active'),
    -> (3, 'Chikkamagaluru Relief Camp', 'Chikkamagaluru', 600, 420, 'Temporary', '9000000003', 'Active'),
    -> (4, 'Sakleshpur Government College', 'Sakleshpur', 400, 250, 'Temporary', '9000000004', 'Active'),
    -> (5, 'Bengaluru Relief Center', 'Whitefield', 800, 650, 'Temporary', '9000000005', 'Active'),
    -> (6, 'Mangaluru Community Center', 'Mangaluru', 700, 300, 'Permanent', '9000000006', 'Active');
Query OK, 6 rows affected (0.06 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

Figure 6.11: INSERT statement for shelter

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM shelter;
+------------+-------------------------------+---------------+----------+-------------------+--------------+------------+--------+
| shelter_id | shelter_name                  | location      | capacity | current_occupancy | shelter_type | contact_no | status |
+------------+-------------------------------+---------------+----------+-------------------+--------------+------------+--------+
|          1 | Kodagu Government School      | Madikeri      |      500 |               320 | Temporary    | 9000000001 | Active |
|          2 | Somwarpet Community Hall      | Somwarpet     |      300 |               180 | Temporary    | 9000000002 | Active |
|          3 | Chikkamagaluru Relief Camp    | Chikkamagaluru |      600 |               420 | Temporary    | 9000000003 | Active |
|          4 | Sakleshpur Government College | Sakleshpur    |      400 |               250 | Temporary    | 9000000004 | Active |
|          5 | Bengaluru Relief Center       | Whitefield    |      800 |               650 | Temporary    | 9000000005 | Active |
|          6 | Mangaluru Community Center    | Mangaluru     |      700 |               300 | Permanent    | 9000000006 | Active |
+------------+-------------------------------+---------------+----------+-------------------+--------------+------------+--------+
6 rows in set (0.00 sec)
```

Figure 6.12: Contents of shelter after insertion

6.7 shelter_allocation table

```
PES2UG24CS457 PES2UG24CS463 >INSERT INTO shelter_allocation
    -> (allocation_id, person_id, shelter_id, allocation_date, status, remarks)
    -> VALUES
    -> (1, 1, 1, '2026-08-20', 'Allocated', 'Family accommodation'),
    -> (2, 2, 1, '2026-08-20', 'Allocated', 'Medical assistance required'),
    -> (3, 3, 2, '2026-08-21', 'Allocated', 'Temporary shelter provided'),
    -> (4, 4, 3, '2026-08-22', 'Allocated', 'Priority case'),
    -> (5, 6, 4, '2026-08-23', 'Allocated', 'Shelter provided'),
    -> (6, 8, 5, '2026-08-24', 'Allocated', 'Medical priority case'),
    -> (7, 9, 5, '2026-08-24', 'Allocated', 'Temporary accommodation'),
    -> (8, 10, 6, '2026-08-25', 'Allocated', 'Family accommodation');
Query OK, 8 rows affected (0.05 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

Figure 6.13: INSERT statement for shelter_allocation

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM shelter_allocation;
+---------------+-----------+------------+-----------------+-----------+-----------------------------+
| allocation_id | person_id | shelter_id | allocation_date | status    | remarks                     |
+---------------+-----------+------------+-----------------+-----------+-----------------------------+
|             1 |         1 |          1 | 2026-08-20      | Allocated | Family accommodation        |
|             2 |         2 |          1 | 2026-08-20      | Allocated | Medical assistance required |
|             3 |         3 |          2 | 2026-08-21      | Allocated | Temporary shelter provided  |
|             4 |         4 |          3 | 2026-08-22      | Allocated | Priority case               |
|             5 |         6 |          4 | 2026-08-23      | Allocated | Shelter provided            |
|             6 |         8 |          5 | 2026-08-24      | Allocated | Medical priority case       |
|             7 |         9 |          5 | 2026-08-24      | Allocated | Temporary accommodation     |
|             8 |        10 |          6 | 2026-08-25      | Allocated | Family accommodation        |
+---------------+-----------+------------+-----------------+-----------+-----------------------------+
8 rows in set (0.00 sec)
```

Figure 6.14: Contents of shelter_allocation after insertion

6.8 relief_request table

```
PES2UG24CS457 PES2UG24CS463 >INSERT INTO relief_request
    -> (request_id, area_id, person_id, resource_id, request_type, quantity_needed, priority, request_date, status, remarks)
    -> VALUES
    -> (1, 1, 1, 4, 'Shelter', 2, 'High', '2026-08-20', 'Approved', 'Shelter required due to landslide'),
    -> (2, 1, 2, 3, 'Medical', 1, 'High', '2026-08-20', 'Completed', 'Medical kit required'),
    -> (3, 2, 3, 1, 'Food', 50, 'Medium', '2026-08-21', 'Approved', 'Food supplies required'),
    -> (4, 3, 4, 3, 'Medical', 2, 'High', '2026-08-22', 'Pending', 'Medical assistance required'),
    -> (5, 4, 6, 4, 'Shelter', 3, 'High', '2026-08-23', 'Approved', 'Blankets required'),
    -> (6, 5, 7, 1, 'Food', 75, 'High', '2026-08-24', 'Completed', 'Food distribution required'),
    -> (7, 7, 8, 3, 'Medical', 2, 'High', '2026-08-24', 'Pending', 'Medical kits required'),
    -> (8, 8, 9, 6, 'Shelter', 4, 'Medium', '2026-08-25', 'Approved', 'Tents required'),
    -> (9, 9, 10, 1, 'Food', 100, 'High', '2026-08-25', 'Approved', 'Food supplies required'),
    -> (10, 9, 10, 2, 'Water', 200, 'Medium', '2026-08-25', 'Pending', 'Drinking water required');
Query OK, 10 rows affected (0.06 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

Figure 6.15: INSERT statement for relief_request

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM relief_request;
+------------+---------+-----------+-------------+--------------+-----------------+----------+--------------+-----------+---------------------------------+
| request_id | area_id | person_id | resource_id | request_type | quantity_needed | priority | request_date | status    | remarks                         |
+------------+---------+-----------+-------------+--------------+-----------------+----------+--------------+-----------+---------------------------------+
|          1 |       1 |         1 |           4 | Shelter      |               2 | High     | 2026-08-20   | Approved  | Shelter required due to landslide |
|          2 |       1 |         2 |           3 | Medical      |               1 | High     | 2026-08-20   | Completed | Medical kit required            |
|          3 |       2 |         3 |           1 | Food         |              50 | Medium   | 2026-08-21   | Approved  | Food supplies required          |
|          4 |       3 |         4 |           3 | Medical      |               2 | High     | 2026-08-22   | Pending   | Medical assistance required     |
|          5 |       4 |         6 |           4 | Shelter      |               3 | High     | 2026-08-23   | Approved  | Blankets required               |
|          6 |       5 |         7 |           1 | Food         |              75 | High     | 2026-08-24   | Completed | Food distribution required      |
|          7 |       7 |         8 |           3 | Medical      |               2 | High     | 2026-08-24   | Pending   | Medical kits required           |
|          8 |       8 |         9 |           6 | Shelter      |               4 | Medium   | 2026-08-25   | Approved  | Tents required                  |
|          9 |       9 |        10 |           1 | Food         |             100 | High     | 2026-08-25   | Approved  | Food supplies required          |
|         10 |       9 |        10 |           2 | Water        |             200 | Medium   | 2026-08-25   | Pending   | Drinking water required         |
+------------+---------+-----------+-------------+--------------+-----------------+----------+--------------+-----------+---------------------------------+
10 rows in set (0.00 sec)
```

Figure 6.16: Contents of relief_request after insertion

6.9 medical_request table

```
PES2UG24CS457 PES2UG24CS463 >INSERT INTO medical_request
    -> (medical_request_id, person_id, medical_issue, priority, request_date, status, assigned_to)
    -> VALUES
    -> (1, 2, 'Injury and dehydration', 'High', '2026-08-20', 'Completed', 'Dr. Ravi'),
    -> (2, 4, 'Leg injury', 'High', '2026-08-22', 'Completed', 'Dr. Anjali'),
    -> (3, 5, 'Fever and weakness', 'Medium', '2026-08-22', 'Pending', 'Dr. Kumar'),
    -> (4, 8, 'Respiratory problem', 'High', '2026-08-24', 'Pending', 'Dr. Priya'),
    -> (5, 10, 'Minor injuries', 'Medium', '2026-08-25', 'Completed', 'Dr. Arun'),
    -> (6, 1, 'Back injury', 'High', '2026-08-20', 'Completed', 'Dr. Ravi'),
    -> (7, 6, 'Fracture', 'High', '2026-08-23', 'Pending', 'Dr. Anjali'),
    -> (8, 9, 'Dehydration', 'Medium', '2026-08-25', 'Completed', 'Dr. Kumar');
Query OK, 8 rows affected (0.06 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

Figure 6.17: INSERT statement for medical_request

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM medical_request;
+--------------------+------------+-----------------------+----------+--------------+-----------+-------------+-------+
| medical_request_id | person_id | medical_issue          | priority | request_date | status    | assigned_to | notes |
+--------------------+------------+-----------------------+----------+--------------+-----------+-------------+-------+
|                  1 |          2 | Injury and dehydration | High     | 2026-08-20   | Completed | Dr. Ravi    | NULL  |
|                  2 |          4 | Leg injury             | High     | 2026-08-22   | Completed | Dr. Anjali  | NULL  |
|                  3 |          5 | Fever and weakness     | Medium   | 2026-08-22   | Pending   | Dr. Kumar   | NULL  |
|                  4 |          8 | Respiratory problem    | High     | 2026-08-24   | Pending   | Dr. Priya   | NULL  |
|                  5 |         10 | Minor injuries         | Medium   | 2026-08-25   | Completed | Dr. Arun    | NULL  |
|                  6 |          1 | Back injury            | High     | 2026-08-20   | Completed | Dr. Ravi    | NULL  |
|                  7 |          6 | Fracture               | High     | 2026-08-23   | Pending   | Dr. Anjali  | NULL  |
|                  8 |          9 | Dehydration            | Medium   | 2026-08-25   | Completed | Dr. Kumar   | NULL  |
+--------------------+------------+-----------------------+----------+--------------+-----------+-------------+-------+
8 rows in set (0.00 sec)
```

Figure 6.18: Contents of medical_request after insertion

6.10 supply_distribution table

```
PES2UG24CS457 PES2UG24CS463 >INSERT INTO supply_distribution
    -> (distribution_id, resource_id, shelter_id, area_id,
    -> quantity_distributed, distribution_date, distributed_by, remarks)
    -> VALUES
    -> (1, 1, 1, 1, 500, '2026-08-20', 'Relief Team A', 'Rice distributed to affected families'),
    -> (2, 2, 1, 1, 1000, '2026-08-20', 'Relief Team A', 'Drinking water distributed'),
    -> (3, 4, 2, 2, 200, '2026-08-21', 'Relief Team B', 'Blankets distributed'),
    -> (4, 3, 3, 3, 50, '2026-08-22', 'Medical Team A', 'Medical kits distributed'),
    -> (5, 6, 4, 5, 100, '2026-08-23', 'Relief Team C', 'Tents distributed'),
    -> (6, 1, 5, 7, 750, '2026-08-24', 'Relief Team D', 'Rice distributed'),
    -> (7, 3, 5, 7, 75, '2026-08-24', 'Medical Team B', 'First aid kits distributed'),
    -> (8, 2, 6, 9, 1500, '2026-08-25', 'Relief Team E', 'Drinking water distributed');
Query OK, 8 rows affected (0.06 sec)
Records: 8  Duplicates: 0  Warnings: 0
```

Figure 6.19: INSERT statement for supply_distribution

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM supply_distribution;
+-----------------+-------------+------------+---------+---------------------+-------------------+---------------+-----------------------------------+
| distribution_id | resource_id | shelter_id | area_id | quantity_distributed | distribution_date | distributed_by | remarks                           |
+-----------------+-------------+------------+---------+---------------------+-------------------+---------------+-----------------------------------+
|               1 |           1 |          1 |       1 |                 500 | 2026-08-20        | Relief Team A  | Rice distributed to affected families |
|               2 |           2 |          1 |       1 |                1000 | 2026-08-20        | Relief Team A  | Drinking water distributed        |
|               3 |           4 |          2 |       2 |                 200 | 2026-08-21        | Relief Team B  | Blankets distributed              |
|               4 |           3 |          3 |       3 |                  50 | 2026-08-22        | Medical Team A | Medical kits distributed          |
|               5 |           6 |          4 |       5 |                 100 | 2026-08-23        | Relief Team C  | Tents distributed                 |
|               6 |           1 |          5 |       7 |                 750 | 2026-08-24        | Relief Team D  | Rice distributed                  |
|               7 |           3 |          5 |       7 |                  75 | 2026-08-24        | Medical Team B | First aid kits distributed        |
|               8 |           2 |          6 |       9 |                1500 | 2026-08-25        | Relief Team E  | Drinking water distributed        |
+-----------------+-------------+------------+---------+---------------------+-------------------+---------------+-----------------------------------+
8 rows in set (0.00 sec)
```

Figure 6.20: Contents of supply_distribution after insertion

6.11 rescue_team table

```
PES2UG24CS457 PES2UG24CS463 >INSERT INTO rescue_team
    -> (team_id, team_name, team_type, team_leader, contact_no, status)
    -> VALUES
    -> (1, 'Kodagu Rescue Team', 'Flood Rescue', 'Vikram Singh', '9100000001', 'Active'),
    -> (2, 'Chikkamagaluru Rescue Team', 'Landslide Rescue', 'Rakesh Kumar', '9100000002', 'Active'),
    -> (3, 'Sakleshpur Rescue Team', 'Landslide Rescue', 'Manoj Rao', '9100000003', 'Active'),
    -> (4, 'Bengaluru Emergency Team', 'Flood Rescue', 'Arun Kumar', '9100000004', 'Active'),
    -> (5, 'Dakshina Kannada Relief Team', 'Flood Relief', 'Suresh Shetty', '9100000005', 'Available');
Query OK, 5 rows affected (0.06 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

Figure 6.21: INSERT statement for rescue_team

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM rescue_team;
+---------+-------------------------------+-----------------+---------------+------------+-----------+
| team_id | team_name                     | team_type       | team_leader   | contact_no | status    |
+---------+-------------------------------+-----------------+---------------+------------+-----------+
|       1 | Kodagu Rescue Team            | Flood Rescue    | Vikram Singh  | 9100000001 | Active    |
|       2 | Chikkamagaluru Rescue Team    | Landslide Rescue| Rakesh Kumar  | 9100000002 | Active    |
|       3 | Sakleshpur Rescue Team        | Landslide Rescue| Manoj Rao     | 9100000003 | Active    |
|       4 | Bengaluru Emergency Team      | Flood Rescue    | Arun Kumar    | 9100000004 | Active    |
|       5 | Dakshina Kannada Relief Team  | Flood Relief    | Suresh Shetty | 9100000005 | Available |
+---------+-------------------------------+-----------------+---------------+------------+-----------+
5 rows in set (0.00 sec)
```

Figure 6.22: Contents of rescue_team after insertion

6.12 vehicle table

```
PES2UG24CS457 PES2UG24CS463 >INSERT INTO vehicle
    -> (vehicle_id, team_id, vehicle_type, registration_no, capacity, status, current_location)
    -> VALUES
    -> (1, 1, 'Rescue Van', 'KA01AB1234', 10, 'Available', 'Madikeri'),
    -> (2, 1, 'Rescue Boat', 'KA02CD5678', 15, 'In Use', 'Kodagu'),
    -> (3, 2, 'Rescue Truck', 'KA03EF9012', 20, 'Available', 'Chikkamagaluru'),
    -> (4, 3, 'Rescue Jeep', 'KA04GH3456', 8, 'In Use', 'Sakleshpur'),
    -> (5, 4, 'Ambulance', 'KA05IJ7890', 6, 'Available', 'Bengaluru'),
    -> (6, 5, 'Supply Truck', 'KA06KL1234', 30, 'Available', 'Mangaluru');
Query OK, 6 rows affected (0.06 sec)
Records: 6  Duplicates: 0  Warnings: 0
```

Figure 6.23: INSERT statement for vehicle

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM vehicle;
+------------+---------+--------------+-----------------+----------+-----------+------------------+
| vehicle_id | team_id | vehicle_type | registration_no | capacity | status    | current_location |
+------------+---------+--------------+-----------------+----------+-----------+------------------+
|          1 |       1 | Rescue Van   | KA01AB1234      |       10 | Available | Madikeri         |
|          2 |       1 | Rescue Boat  | KA02CD5678      |       15 | In Use    | Kodagu           |
|          3 |       2 | Rescue Truck | KA03EF9012      |       20 | Available | Chikkamagaluru   |
|          4 |       3 | Rescue Jeep  | KA04GH3456      |        8 | In Use    | Sakleshpur       |
|          5 |       4 | Ambulance    | KA05IJ7890      |        6 | Available | Bengaluru        |
|          6 |       5 | Supply Truck | KA06KL1234      |       30 | Available | Mangaluru        |
+------------+---------+--------------+-----------------+----------+-----------+------------------+
6 rows in set (0.00 sec)
```

Figure 6.24: Contents of vehicle after insertion

7. QUERIES

7.1 Simple Query with GROUP BY and Aggregate Functions

SELECT

Basic SELECT queries retrieve selected columns and filter rows with WHERE.

```
PES2UG24CS457 PES2UG24CS463 >SELECT disaster_id, disaster_name, disaster_type, status
    -> FROM disaster;
+-------------+-------------------------+---------------+--------+
| disaster_id | disaster_name           | disaster_type | status |
+-------------+-------------------------+---------------+--------+
|           1 | Kodagu landslide        | Landslide     | Active |
|           2 | chikamangluru landslide | Landslide     | Active |
|           3 | Saklespura landslide    | Landslide     | closed |
|           4 |  Bengluru urban flood   | Flood         | Active |
|           5 |  Dakshina kannadaflood  | Flood         | closed |
+-------------+-------------------------+---------------+--------+
5 rows in set (0.00 sec)
```

Figure 7.1: SELECT specific columns from disaster

```
PES2UG24CS457 PES2UG24CS463 >SELECT area_id, area_name, location, people_affected
    -> FROM affected_area
    -> WHERE priority_level = 'High';
+---------+----------------+------------------+-----------------+
| area_id | area_name      | location         | people_affected |
+---------+----------------+------------------+-----------------+
|       1 | Madikeri       | Kodagu           |             900 |
|       3 | Chikkamagaluru | Chikkamagaluru   |            1200 |
|       5 | Sakleshpur     | Hassan           |             500 |
|       7 | Whitefield     | Bengaluru        |            1200 |
|       8 | Yelahanka      | Bengaluru        |             750 |
|       9 | Mangaluru      | Dakshina Kannada |            1500 |
+---------+----------------+------------------+-----------------+
6 rows in set (0.00 sec)
```

Figure 7.2: SELECT affected areas with High priority

```
PES2UG24CS457 PES2UG24CS463 >SELECT resource_id, resource_name, resource_type, unit
    -> FROM resource;
+-------------+----------------+---------------+--------+
| resource_id | resource_name  | resource_type | unit   |
+-------------+----------------+---------------+--------+
|           1 | Rice           | Food          | Kg     |
|           2 | Drinking Water | Water         | Litres |
|           3 | Medical Kit    | Medical       | Kits   |
|           4 | Blankets       | Shelter       | Pieces |
|           5 | First Aid Kit  | Medical       | Kits   |
|           6 | Tents          | Shelter       | Pieces |
+-------------+----------------+---------------+--------+
6 rows in set (0.00 sec)
```

Figure 7.3: SELECT from resource

ORDER BY

```
PES2UG24CS457 PES2UG24CS463 >SELECT team_id, team_name, team_type, status
    -> FROM rescue_team
    -> ORDER BY team_name ASC;
+---------+-------------------------------+------------------+--------+
| team_id | team_name                     | team_type        | status |
+---------+-------------------------------+------------------+--------+
|       4 | Bengaluru Emergency Team      | Flood Rescue     | Active |
|       2 | Chikkamagaluru Rescue Team    | Landslide Rescue | Active |
|       5 | Dakshina Kannada Relief Team  | Flood Relief     | Active |
|       1 | Kodagu Rescue Team            | Flood Rescue     | Active |
|       3 | Sakleshpur Rescue Team        | Landslide Rescue | Active |
+---------+-------------------------------+------------------+--------+
5 rows in set (0.00 sec)
```

Figure 7.4: Rescue teams ordered by team_name (ascending)

```
PES2UG24CS457 PES2UG24CS463 >SELECT allocation_id, person_id, shelter_id, allocation_date, status
    -> FROM shelter_allocation
    -> ORDER BY allocation_date DESC;
+---------------+-----------+------------+-----------------+-----------+
| allocation_id | person_id | shelter_id | allocation_date | status    |
+---------------+-----------+------------+-----------------+-----------+
|             8 |        10 |          6 | 2026-08-25      | Allocated |
|             6 |         8 |          5 | 2026-08-24      | Allocated |
|             7 |         9 |          5 | 2026-08-24      | Allocated |
|             5 |         6 |          4 | 2026-08-23      | Allocated |
|             4 |         4 |          3 | 2026-08-22      | Allocated |
|             3 |         3 |          2 | 2026-08-21      | Allocated |
|             1 |         1 |          1 | 2026-08-20      | Allocated |
|             2 |         2 |          1 | 2026-08-20      | Allocated |
+---------------+-----------+------------+-----------------+-----------+
8 rows in set (0.00 sec)
```

Figure 7.5: Shelter allocations ordered by allocation_date (descending)

GROUP BY

```
PES2UG24CS457 PES2UG24CS463 >SELECT request_type, COUNT(*) AS total_requests
    -> FROM relief_request
    -> GROUP BY request_type;
+--------------+----------------+
| request_type | total_requests |
+--------------+----------------+
| Shelter      |              3 |
| Medical      |              3 |
| Food         |              3 |
+--------------+----------------+
3 rows in set (0.02 sec)
```

Figure 7.6: Number of relief requests for each request_type

```
PES2UG24CS457 PES2UG24CS463 >SELECT status, COUNT(*) AS total_requests
    -> FROM medical_request
    -> GROUP BY status;
+-----------+----------------+
| status    | total_requests |
+-----------+----------------+
| Completed |              4 |
| Pending   |              3 |
+-----------+----------------+
2 rows in set (0.01 sec)
```

Figure 7.7: Number of medical requests for each status

HAVING

```
PES2UG24CS457 PES2UG24CS463 >SELECT distributed_by, COUNT(*) AS total_distributions
    -> FROM supply_distribution
    -> GROUP BY distributed_by
    -> HAVING COUNT(*) > 1;
+----------------+---------------------+
| distributed_by | total_distributions |
+----------------+---------------------+
| Relief Team A  |                   2 |
+----------------+---------------------+
1 row in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >
```

Figure 7.8: Distributors with more than one distribution (HAVING COUNT(*) > 1)

```
PES2UG24CS457 PES2UG24CS463 >SELECT priority_level, COUNT(*) AS total_areas
    -> FROM affected_area
    -> GROUP BY priority_level
    -> HAVING COUNT(*) > 2;
+----------------+-------------+
| priority_level | total_areas |
+----------------+-------------+
| High           |           6 |
| Medium         |           4 |
+----------------+-------------+
2 rows in set (0.01 sec)
```

Figure 7.9: Priority levels having more than two affected areas

Aggregate functions – SUM, COUNT, AVG, MAX, MIN

```
PES2UG24CS457 PES2UG24CS463 >SELECT SUM(quantity_available) AS total_quantity
    -> FROM inventory;
+----------------+
| total_quantity |
+----------------+
|          19100 |
+----------------+
1 row in set (0.00 sec)
```

Figure 7.10: SUM of quantity_available in inventory

```
1 row in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >SELECT SUM(quantity_distributed) AS total_quantity_distributed
    -> FROM supply_distribution;
+---------------------------+
| total_quantity_distributed |
+---------------------------+
|                      2675 |
+---------------------------+
1 row in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >
```

Figure 7.11: SUM of quantity_distributed in supply_distribution

```
PES2UG24CS457 PES2UG24CS463 >SELECT COUNT(*) AS total_affected_persons
    -> FROM affected_person;
+------------------------+
| total_affected_persons |
+------------------------+
|                     10 |
+------------------------+
1 row in set (0.01 sec)
```

Figure 7.12: COUNT of affected persons

```
1 row in set (0.01 sec)

PES2UG24CS457 PES2UG24CS463 >SELECT COUNT(*) AS total_rescue_teams
    -> FROM rescue_team;
+--------------------+
| total_rescue_teams |
+--------------------+
|                  5 |
+--------------------+
1 row in set (0.00 sec)
```

Figure 7.13: COUNT of rescue teams

```
PES2UG24CS457 PES2UG24CS463 >SELECT AVG(capacity) AS average_capacity
    -> FROM shelter;
+------------------+
| average_capacity |
+------------------+
|         550.0000 |
+------------------+
1 row in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >
```

Figure 7.14: AVG shelter capacity

```
PES2UG24CS457 PES2UG24CS463 >SELECT AVG(age) AS average_age
    -> FROM affected_person;
+-------------+
| average_age |
+-------------+
|     43.1000 |
+-------------+
1 row in set (0.00 sec)
```

Figure 7.15: AVG age of affected persons

```
PES2UG24CS457 PES2UG24CS463 >SELECT MAX(quantity_available) AS maximum_quantity
    -> FROM inventory;
+------------------+
| maximum_quantity |
+------------------+
|            10000 |
+------------------+
1 row in set (0.01 sec)

PES2UG24CS457 PES2UG24CS463 >
```

Figure 7.16: MAX quantity_available in inventory

```
PES2UG24CS457 PES2UG24CS463 >SELECT MAX(people_affected) AS maximum_people_affected
    -> FROM affected_area;
+------------------------+
| maximum_people_affected |
+------------------------+
|                   1500 |
+------------------------+
1 row in set (0.00 sec)
```

Figure 7.17: MAX people_affected in affected_area

```
PES2UG24CS457 PES2UG24CS463 >SELECT MIN(quantity_available) AS minimum_quantity
    -> FROM inventory;
+------------------+
| minimum_quantity |
+------------------+
|              300 |
+------------------+
1 row in set (0.00 sec)
```

Figure 7.18: MIN quantity_available in inventory

```
PES2UG24CS457 PES2UG24CS463 >SELECT MIN(capacity) AS minimum_capacity
    -> FROM shelter;
+------------------+
| minimum_capacity |
+------------------+
|              300 |
+------------------+
1 row in set (0.00 sec)
```

Figure 7.19: MIN shelter capacity

JOIN queries

```
PES2UG24CS457 PES2UG24CS463 >SELECT
    ->     a.area_id,
    ->     a.area_name,
    ->     a.location,
    ->     d.disaster_name,
    ->     d.disaster_type
    -> FROM affected_area a
    -> JOIN disaster d
    -> ON a.disaster_id = d.disaster_id;
+---------+----------------+------------------+-------------------------+---------------+
| area_id | area_name      | location         | disaster_name           | disaster_type |
+---------+----------------+------------------+-------------------------+---------------+
|       1 | Madikeri       | Kodagu           | Kodagu landslide        | Landslide     |
|       2 | Somwarpet      | Kodagu           | Kodagu landslide        | Landslide     |
|       3 | Chikkamagaluru | Chikkamagaluru   | chikamangluru landslide | Landslide     |
|       4 | Koppa          | Chikkamagaluru   | chikamangluru landslide | Landslide     |
|       5 | Sakleshpur     | Hassan           | Saklespura landslide    | Landslide     |
|       6 | Belur          | Hassan           | Saklespura landslide    | Landslide     |
|       7 | Whitefield     | Bengaluru        |  Bengluru urban flood   | Flood         |
|       8 | Yelahanka      | Bengaluru        |  Bengluru urban flood   | Flood         |
|       9 | Mangaluru      | Dakshina Kannada |  Dakshina kannadaflood  | Flood         |
|      10 | Udupi Road     | Dakshina Kannada |  Dakshina kannadaflood  | Flood         |
+---------+----------------+------------------+-------------------------+---------------+
10 rows in set (0.00 sec)
```

Figure 7.20: INNER JOIN – affected_area with disaster

```
PES2UG24CS457 PES2UG24CS463 >SELECT
    ->     sa.allocation_id,
    ->     sa.person_id,
    ->     s.shelter_name,
    ->     s.location,
    ->     sa.allocation_date,
    ->     sa.status
    -> FROM shelter_allocation sa
    -> JOIN shelter s
    -> ON sa.shelter_id = s.shelter_id;
+---------------+-----------+-------------------------------+----------------+-----------------+-----------+
| allocation_id | person_id | shelter_name                  | location       | allocation_date | status    |
+---------------+-----------+-------------------------------+----------------+-----------------+-----------+
|             1 |         1 | Kodagu Government School      | Madikeri       | 2026-08-20      | Allocated |
|             2 |         2 | Kodagu Government School      | Madikeri       | 2026-08-20      | Allocated |
|             3 |         3 | Somwarpet Community Hall      | Somwarpet      | 2026-08-21      | Allocated |
|             4 |         4 | Chikkamagaluru Relief Camp    | Chikkamagaluru | 2026-08-22      | Allocated |
|             5 |         6 | Sakleshpur Government College | Sakleshpur     | 2026-08-23      | Allocated |
|             6 |         8 | Bengaluru Relief Center       | Whitefield     | 2026-08-24      | Allocated |
|             7 |         9 | Bengaluru Relief Center       | Whitefield     | 2026-08-24      | Allocated |
|             8 |        10 | Mangaluru Community Center    | Mangaluru      | 2026-08-25      | Allocated |
+---------------+-----------+-------------------------------+----------------+-----------------+-----------+
8 rows in set (0.00 sec)
```

Figure 7.21: INNER JOIN – shelter_allocation with shelter

```
PES2UG24CS457 PES2UG24CS463 >SELECT
    ->     a.area_id,
    ->     a.area_name,
    ->     d.disaster_name,
    ->     d.disaster_type
    -> FROM affected_area a
    -> LEFT JOIN disaster d
    -> ON a.disaster_id = d.disaster_id;
+---------+---------------+-------------------------+---------------+
| area_id | area_name     | disaster_name           | disaster_type |
+---------+---------------+-------------------------+---------------+
|       1 | Madikeri      | Kodagu landslide        | Landslide     |
|       2 | Somwarpet     | Kodagu landslide        | Landslide     |
|       3 | Chikkamagaluru | chikamangluru landslide | Landslide     |
|       4 | Koppa         | chikamangluru landslide | Landslide     |
|       5 | Sakleshpur    | Saklespura landslide    | Landslide     |
|       6 | Belur         | Saklespura landslide    | Landslide     |
|       7 | Whitefield    |  Bengluru urban flood   | Flood         |
|       8 | Yelahanka     |  Bengluru urban flood   | Flood         |
|       9 | Mangaluru     |  Dakshina kannadaflood  | Flood         |
|      10 | Udupi Road    |  Dakshina kannadaflood  | Flood         |
+---------+---------------+-------------------------+---------------+
10 rows in set (0.00 sec)
```

Figure 7.22: LEFT JOIN – affected_area with disaster

```
PES2UG24CS457 PES2UG24CS463 >SELECT
    ->     d.disaster_id,
    ->     d.disaster_name,
    ->     a.area_name,
    ->     a.people_affected
    -> FROM disaster d
    -> RIGHT JOIN affected_area a
    -> ON d.disaster_id = a.disaster_id;
+-------------+-------------------------+---------------+-----------------+
| disaster_id | disaster_name           | area_name     | people_affected |
+-------------+-------------------------+---------------+-----------------+
|           1 | Kodagu landslide        | Madikeri      |             900 |
|           1 | Kodagu landslide        | Somwarpet     |             600 |
|           2 | chikamangluru landslide | Chikkamagaluru |            1200 |
|           2 | chikamangluru landslide | Koppa         |             700 |
|           3 | Saklespura landslide    | Sakleshpur    |             500 |
|           3 | Saklespura landslide    | Belur         |             800 |
|           4 |  Bengluru urban flood   | Whitefield    |            1200 |
|           4 |  Bengluru urban flood   | Yelahanka     |             750 |
|           5 |  Dakshina kannadaflood  | Mangaluru     |            1500 |
|           5 |  Dakshina kannadaflood  | Udupi Road    |             650 |
+-------------+-------------------------+---------------+-----------------+
10 rows in set (0.00 sec)
```

Figure 7.23: RIGHT JOIN – disaster with affected_area

7.2 Update Operation

UPDATE statements change existing data, for example refreshing stock, occupancy, team status and a person's priority.

```
PES2UG24CS457 PES2UG24CS463 >UPDATE inventory
    -> SET quantity_available = 5500
    -> WHERE inventory_id = 1;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >SELECT * FROM inventory
    -> WHERE inventory_id = 1;
+--------------+-------------+--------------------+--------------+----------------------+-------------------+
| inventory_id | resource_id | quantity_available | last_updated | location             | storage_condition |
+--------------+-------------+--------------------+--------------+----------------------+-------------------+
|            1 |           1 |               5500 | 2026-08-25   | Kodagu Relief Center | Good              |
+--------------+-------------+--------------------+--------------+----------------------+-------------------+
1 row in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >
```

Figure 7.24: UPDATE inventory – quantity_available set to 5500 for inventory_id 1

```
PES2UG24CS457 PES2UG24CS463 >UPDATE shelter
    -> SET current_occupancy = 350
    -> WHERE shelter_id = 1;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >SELECT * FROM shelter
    -> WHERE shelter_id = 1;
+------------+-------------------------+----------+----------+-------------------+--------------+------------+--------+----------------+
| shelter_id | shelter_name            | location | capacity | current_occupancy | shelter_type | contact_no | status | last_inspected |
+------------+-------------------------+----------+----------+-------------------+--------------+------------+--------+----------------+
|          1 | Kodagu Government School | Madikeri |      500 |               350 | Temporary    | 9000000001 | Active | 2026-08-27     |
+------------+-------------------------+----------+----------+-------------------+--------------+------------+--------+----------------+
1 row in set (0.00 sec)
```

Figure 7.25: UPDATE shelter – current_occupancy set to 350 for shelter_id 1

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM rescue_team
    -> WHERE team_id = 5;
+---------+-------------------------------+--------------+---------------+------------+-----------+
| team_id | team_name                     | team_type    | team_leader   | contact_no | status    |
+---------+-------------------------------+--------------+---------------+------------+-----------+
|       5 | Dakshina Kannada Relief Team | Flood Relief | Suresh Shetty | 9100000005 | Available |
+---------+-------------------------------+--------------+---------------+------------+-----------+
1 row in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >UPDATE rescue_team
    -> SET status = 'Active'
    -> WHERE team_id = 5;
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >SELECT * FROM rescue_team
    -> WHERE team_id = 5;
+---------+-------------------------------+--------------+---------------+------------+--------+
| team_id | team_name                     | team_type    | team_leader   | contact_no | status |
+---------+-------------------------------+--------------+---------------+------------+--------+
|       5 | Dakshina Kannada Relief Team | Flood Relief | Suresh Shetty | 9100000005 | Active |
+---------+-------------------------------+--------------+---------------+------------+--------+
1 row in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >
```

Figure 7.26: UPDATE rescue_team – status changed from Available to Active

```
PES2UG24CS457 PES2UG24CS463 >SELECT * FROM affected_person
    -> WHERE person_id = 10;
+-----------+---------+---------------+-----+--------+------------+-----------------------------+-----------+----------+--------------------+
| person_id | area_id | person_name   | age | gender | contact_no | address                     | need_type | priority | emergency_contact  |
+-----------+---------+---------------+-----+--------+------------+-----------------------------+-----------+----------+--------------------+
|        10 |       9 | Mohammed Irfan |  56 | Male   | 9876543210 | Mangaluru, Dakshina Kannada | Food      | High     | 9876543210         |
+-----------+---------+---------------+-----+--------+------------+-----------------------------+-----------+----------+--------------------+
1 row in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >UPDATE affected_person
    -> SET priority = 'Medium'
    -> WHERE person_id = 10;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >SELECT * FROM affected_person
    -> WHERE person_id = 10;
+-----------+---------+---------------+-----+--------+------------+-----------------------------+-----------+----------+--------------------+
| person_id | area_id | person_name   | age | gender | contact_no | address                     | need_type | priority | emergency_contact  |
+-----------+---------+---------------+-----+--------+------------+-----------------------------+-----------+----------+--------------------+
|        10 |       9 | Mohammed Irfan |  56 | Male   | 9876543210 | Mangaluru, Dakshina Kannada | Food      | Medium   | 9876543210         |
+-----------+---------+---------------+-----+--------+------------+-----------------------------+-----------+----------+--------------------+
1 row in set (0.00 sec)
```

Figure 7.27: UPDATE affected_person – priority changed from High to Medium for person_id 10

7.3 Delete Operation

DELETE statements remove specific rows using a WHERE condition.

```
PES2UG24CS457 PES2UG24CS463 >DELETE FROM vehicle
    -> WHERE vehicle_id = 6;
Query OK, 1 row affected (0.04 sec)

PES2UG24CS457 PES2UG24CS463 >select * from vehical;
ERROR 1146 (42S02): Table 'disaster_relief_management.vehical' doesn't exist
PES2UG24CS457 PES2UG24CS463 >select * from vehicle;
+------------+---------+--------------+-----------------+----------+-----------+------------------+
| vehicle_id | team_id | vehicle_type | registration_no | capacity | status    | current_location |
+------------+---------+--------------+-----------------+----------+-----------+------------------+
|          1 |       1 | Rescue Van   | KA01AB1234      |       10 | Available | Madikeri         |
|          2 |       1 | Rescue Boat  | KA02CD5678      |       15 | In Use    | Kodagu           |
|          3 |       2 | Rescue Truck | KA03EF9012      |       20 | Available | Chikkamagaluru   |
|          4 |       3 | Rescue Jeep  | KA04GH3456      |        8 | In Use    | Sakleshpur       |
|          5 |       4 | Ambulance    | KA05IJ7890      |        6 | Available | Bengaluru        |
+------------+---------+--------------+-----------------+----------+-----------+------------------+
5 rows in set (0.00 sec)
```

Figure 7.28: DELETE FROM vehicle WHERE vehicle_id = 6

```
PES2UG24CS457 PES2UG24CS463 >DELETE FROM medical_request
    -> WHERE medical_request_id = 8;
Query OK, 1 row affected (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >select * from medical_request;
+--------------------+-----------+-------------------------+----------+--------------+-----------+-------------+-------+
| medical_request_id | person_id | medical_issue           | priority | request_date | status    | assigned_to | notes |
+--------------------+-----------+-------------------------+----------+--------------+-----------+-------------+-------+
|                  1 |         2 | Injury and dehydration  | High     | 2026-08-20   | Completed | Dr. Ravi    | NULL  |
|                  2 |         4 | Leg injury              | High     | 2026-08-22   | Completed | Dr. Anjali  | NULL  |
|                  3 |         5 | Fever and weakness      | Medium   | 2026-08-22   | Pending   | Dr. Kumar   | NULL  |
|                  4 |         8 | Respiratory problem     | High     | 2026-08-24   | Pending   | Dr. Priya   | NULL  |
|                  5 |        10 | Minor injuries          | Medium   | 2026-08-25   | Completed | Dr. Arun    | NULL  |
|                  6 |         1 | Back injury             | High     | 2026-08-20   | Completed | Dr. Ravi    | NULL  |
|                  7 |         6 | Fracture                | High     | 2026-08-23   | Pending   | Dr. Anjali  | NULL  |
+--------------------+-----------+-------------------------+----------+--------------+-----------+-------------+-------+
7 rows in set (0.00 sec)
```

Figure 7.29: DELETE FROM medical_request WHERE medical_request_id = 8

```
PES2UG24CS457 PES2UG24CS463 >DELETE FROM relief_request
    -> WHERE request_id = 10;
Query OK, 1 row affected (0.04 sec)

PES2UG24CS457 PES2UG24CS463 >SELECT * FROM relief_request;
+------------+---------+-----------+-------------+--------------+-----------------+----------+--------------+-----------+-----------------------------------+
| request_id | area_id | person_id | resource_id | request_type | quantity_needed | priority | request_date | status    | remarks                           |
+------------+---------+-----------+-------------+--------------+-----------------+----------+--------------+-----------+-----------------------------------+
|          1 |       1 |         1 |           4 | Shelter      |               2 | High     | 2026-08-20   | Approved  | Shelter required due to landslide |
|          2 |       1 |         2 |           3 | Medical      |               1 | High     | 2026-08-20   | Completed | Medical kit required              |
|          3 |       2 |         3 |           1 | Food         |              50 | Medium   | 2026-08-21   | Approved  | Food supplies required            |
|          4 |       3 |         4 |           3 | Medical      |               2 | High     | 2026-08-22   | Pending   | Medical assistance required       |
|          5 |       4 |         6 |           4 | Shelter      |               3 | High     | 2026-08-23   | Approved  | Blankets required                 |
|          6 |       5 |         7 |           1 | Food         |              75 | High     | 2026-08-24   | Completed | Food distribution required        |
|          7 |       7 |         8 |           3 | Medical      |               2 | High     | 2026-08-24   | Pending   | Medical kits required             |
|          8 |       8 |         9 |           6 | Shelter      |               4 | Medium   | 2026-08-25   | Approved  | Tents required                    |
|          9 |       9 |        10 |           1 | Food         |             100 | High     | 2026-08-25   | Approved  | Food supplies required            |
+------------+---------+-----------+-------------+--------------+-----------------+----------+--------------+-----------+-----------------------------------+
9 rows in set (0.00 sec)
```

Figure 7.30: DELETE FROM relief_request WHERE request_id = 10

```
PES2UG24CS457 PES2UG24CS463 >DELETE FROM supply_distribution
    -> WHERE distribution_id = 8;
Query OK, 1 row affected (0.05 sec)

PES2UG24CS457 PES2UG24CS463 >SELECT * FROM supply_distribution;
+-----------------+-------------+------------+---------+----------------------+-------------------+---------------+-----------------------------------+
| distribution_id | resource_id | shelter_id | area_id | quantity_distributed | distribution_date | distributed_by | remarks                           |
+-----------------+-------------+------------+---------+----------------------+-------------------+---------------+-----------------------------------+
|               1 |           1 |          1 |       1 |                  500 | 2026-08-20        | Relief Team A  | Rice distributed to affected families |
|               2 |           2 |          1 |       1 |                 1000 | 2026-08-20        | Relief Team A  | Drinking water distributed        |
|               3 |           4 |          2 |       2 |                  200 | 2026-08-21        | Relief Team B  | Blankets distributed              |
|               4 |           3 |          3 |       3 |                   50 | 2026-08-22        | Medical Team A | Medical kits distributed          |
|               5 |           6 |          4 |       5 |                  100 | 2026-08-23        | Relief Team C  | Tents distributed                 |
|               6 |           1 |          5 |       7 |                  750 | 2026-08-24        | Relief Team D  | Rice distributed                  |
|               7 |           3 |          5 |       7 |                   75 | 2026-08-24        | Medical Team B | First aid kits distributed        |
+-----------------+-------------+------------+---------+----------------------+-------------------+---------------+-----------------------------------+
7 rows in set (0.00 sec)
```

Figure 7.31: DELETE FROM supply_distribution WHERE distribution_id = 8

7.4 Correlated Query

A correlated query uses a subquery that refers to a column of the outer query, so the subquery is evaluated once for each outer row. The query below lists shelters that are more crowded than the average occupancy of shelters of the same type.

```
SELECT s.shelter_id, s.shelter_name, s.shelter_type, s.current_occupancy
FROM shelter s
WHERE s.current_occupancy > (
        SELECT AVG(s2.current_occupancy)
        FROM shelter s2
        WHERE s2.shelter_type = s.shelter_type);
```

7.5 Nested Query

A nested query (subquery) places one SELECT inside another. Here the inner query computes the average and the outer query returns rows above that average.

```
PES2UG24CS457 PES2UG24CS463 >SELECT shelter_id, shelter_name, capacity
    -> FROM shelter
    -> WHERE capacity > (
    ->     SELECT AVG(capacity)
    ->     FROM shelter
    -> );
+------------+---------------------------+----------+
| shelter_id | shelter_name              | capacity |
+------------+---------------------------+----------+
|          3 | Chikkamagaluru Relief Camp |      600 |
|          5 | Bengaluru Relief Center    |      800 |
|          6 | Mangaluru Community Center |      700 |
+------------+---------------------------+----------+
3 rows in set (0.01 sec)
```

Figure 7.32: Nested query – shelters with capacity above the average capacity

```
PES2UG24CS457 PES2UG24CS463 >SELECT area_id, area_name, people_affected
    -> FROM affected_area
    -> WHERE people_affected > (
    ->     SELECT AVG(people_affected)
    ->     FROM affected_area
    -> );
+---------+----------------+-----------------+
| area_id | area_name      | people_affected |
+---------+----------------+-----------------+
|       1 | Madikeri       |             900 |
|       3 | Chikkamagaluru |            1200 |
|       7 | Whitefield     |            1200 |
|       9 | Mangaluru      |            1500 |
+---------+----------------+-----------------+
4 rows in set (0.00 sec)
```

Figure 7.33: Nested query – affected areas with people_affected above the average

8. STORED PROCEDURES, FUNCTIONS AND TRIGGERS

8.1 Stored Procedures or Functions

Function – get_people_affected(a_id)

The function takes an area_id and returns the number of people affected in that area. For area 1 (Madikeri) it returns 900.

```
PES2UG24CS457 PES2UG24CS463>CREATE FUNCTION get_people_affected(a_id INT)
    -> RETURNS INT
    -> DETERMINISTIC
    -> BEGIN
    ->     DECLARE total_people INT;
    ->
    ->     SELECT people_affected
    ->     INTO total_people
    ->     FROM affected_area
    ->     WHERE area_id = a_id;
    ->
    ->     RETURN total_people;
    -> END //
Query OK, 0 rows affected (0.09 sec)

PES2UG24CS457 PES2UG24CS463>
PES2UG24CS457 PES2UG24CS463>DELIMITER ;
PES2UG24CS457 PES2UG24CS463>SELECT get_people_affected(1) AS people_affected;
+-----------------+
| people_affected |
+-----------------+
|             900 |
+-----------------+
1 row in set (0.14 sec)

PES2UG24CS457 PES2UG24CS463>
```

Figure 8.1: Function get_people_affected and its output

Procedure – get_active_teams()

The procedure lists all rescue teams whose status is Active.

```
PES2UG24CS457 PES2UG24CS463>DELIMITER //
PES2UG24CS457 PES2UG24CS463>
PES2UG24CS457 PES2UG24CS463>CREATE PROCEDURE get_active_teams()
    -> BEGIN
    ->     SELECT team_id, team_name, team_type, status
    ->     FROM rescue_team
    ->     WHERE status = 'Active';
    -> END //
Query OK, 0 rows affected (0.05 sec)

PES2UG24CS457 PES2UG24CS463>
PES2UG24CS457 PES2UG24CS463>DELIMITER ;
PES2UG24CS457 PES2UG24CS463>CALL get_active_teams();
+---------+-------------------------------+-------------------+--------+
| team_id | team_name                     | team_type         | status |
+---------+-------------------------------+-------------------+--------+
|       1 | Kodagu Rescue Team            | Flood Rescue      | Active |
|       2 | Chikkamagaluru Rescue Team    | Landslide Rescue  | Active |
|       3 | Sakleshpur Rescue Team        | Landslide Rescue  | Active |
|       4 | Bengaluru Emergency Team      | Flood Rescue      | Active |
|       5 | Dakshina Kannada Relief Team  | Flood Relief      | Active |
+---------+-------------------------------+-------------------+--------+
5 rows in set (0.02 sec)

Query OK, 0 rows affected (0.05 sec)

PES2UG24CS457 PES2UG24CS463>
```

Figure 8.2: Procedure get_active_teams and CALL output

8.2 Triggers

Trigger – update_inventory_date

A BEFORE UPDATE trigger on inventory sets last_updated to the current date whenever a stock row is updated.

```
PES2UG24CS457 PES2UG24CS463 >DELIMITER //
PES2UG24CS457 PES2UG24CS463 >
PES2UG24CS457 PES2UG24CS463 >CREATE TRIGGER update_inventory_date
    -> BEFORE UPDATE ON inventory
    -> FOR EACH ROW
    -> BEGIN
    ->     SET NEW.last_updated = CURDATE();
    -> END //
Query OK, 0 rows affected (0.02 sec)

PES2UG24CS457 PES2UG24CS463 >
PES2UG24CS457 PES2UG24CS463 >DELIMITER ;
PES2UG24CS457 PES2UG24CS463 >UPDATE inventory
    -> SET quantity_available = 5600
    -> WHERE inventory_id = 1;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >SELECT inventory_id, quantity_available, last_updated
    -> FROM inventory
    -> WHERE inventory_id = 1;
+--------------+--------------------+--------------+
| inventory_id | quantity_available | last_updated |
+--------------+--------------------+--------------+
|            1 |               5600 | 2026-10-01   |
+--------------+--------------------+--------------+
1 row in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >SHOW TRIGGERS LIKE 'inventory';
+-----------------------+--------+-----------+-----------------------------------------------------------------------+--------+---------------------+-----------------------------------------------------------------------
```

Trigger – update_shelter_status

A BEFORE UPDATE trigger on shelter sets status to “Full” when current_occupancy reaches capacity. Setting occupancy of Sommarpet Community Hall to 300 (its capacity) changed the status to Full.

```
PES2UG24CS457 PES2UG24CS463 >DELIMITER //
PES2UG24CS457 PES2UG24CS463 >
PES2UG24CS457 PES2UG24CS463 >CREATE TRIGGER update_shelter_status
    -> BEFORE UPDATE ON shelter
    -> FOR EACH ROW
    -> BEGIN
    ->     IF NEW.current_occupancy >= NEW.capacity THEN
    ->         SET NEW.status = 'Full';
    ->     END IF;
    -> END //
Query OK, 0 rows affected (0.02 sec)

PES2UG24CS457 PES2UG24CS463 >
PES2UG24CS457 PES2UG24CS463 >DELIMITER ;
PES2UG24CS457 PES2UG24CS463 >UPDATE shelter
    -> SET current_occupancy = 300
    -> WHERE shelter_id = 2;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >SELECT shelter_id, shelter_name, capacity, current_occupancy, status
    -> FROM shelter
    -> WHERE shelter_id = 2;
+------------+--------------------------+----------+-------------------+--------+
| shelter_id | shelter_name             | capacity | current_occupancy | status |
+------------+--------------------------+----------+-------------------+--------+
|          2 | Somwarpet Community Hall |      300 |               300 | Full   |
+------------+--------------------------+----------+-------------------+--------+
1 row in set (0.00 sec)

PES2UG24CS457 PES2UG24CS463 >UPDATE shelter
    -> SET current_occupancy = 180,
    ->     status = 'Active'
    -> WHERE shelter_id = 2;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

PES2UG24CS457 PES2UG24CS463 >SELECT shelter_id, shelter_name, capacity, current_occupancy, status
    -> FROM shelter
    -> WHERE shelter_id = 2;
+------------+--------------------------+----------+-------------------+--------+
| shelter_id | shelter_name             | capacity | current_occupancy | status |
+------------+--------------------------+----------+-------------------+--------+
|          2 | Somwarpet Community Hall |      300 |               180 | Active |
+------------+--------------------------+----------+-------------------+--------+
1 row in set (0.00 sec)
```

Figure 8.4: Trigger update_shelter_status on shelter

9. FRONT END DEVELOPMENT

The project was implemented and demonstrated through the MySQL command-line interface, using SQL statements for all operations shown in the previous chapters.

Figure 9.1: Sign in page – an existing user signs in with email and password.

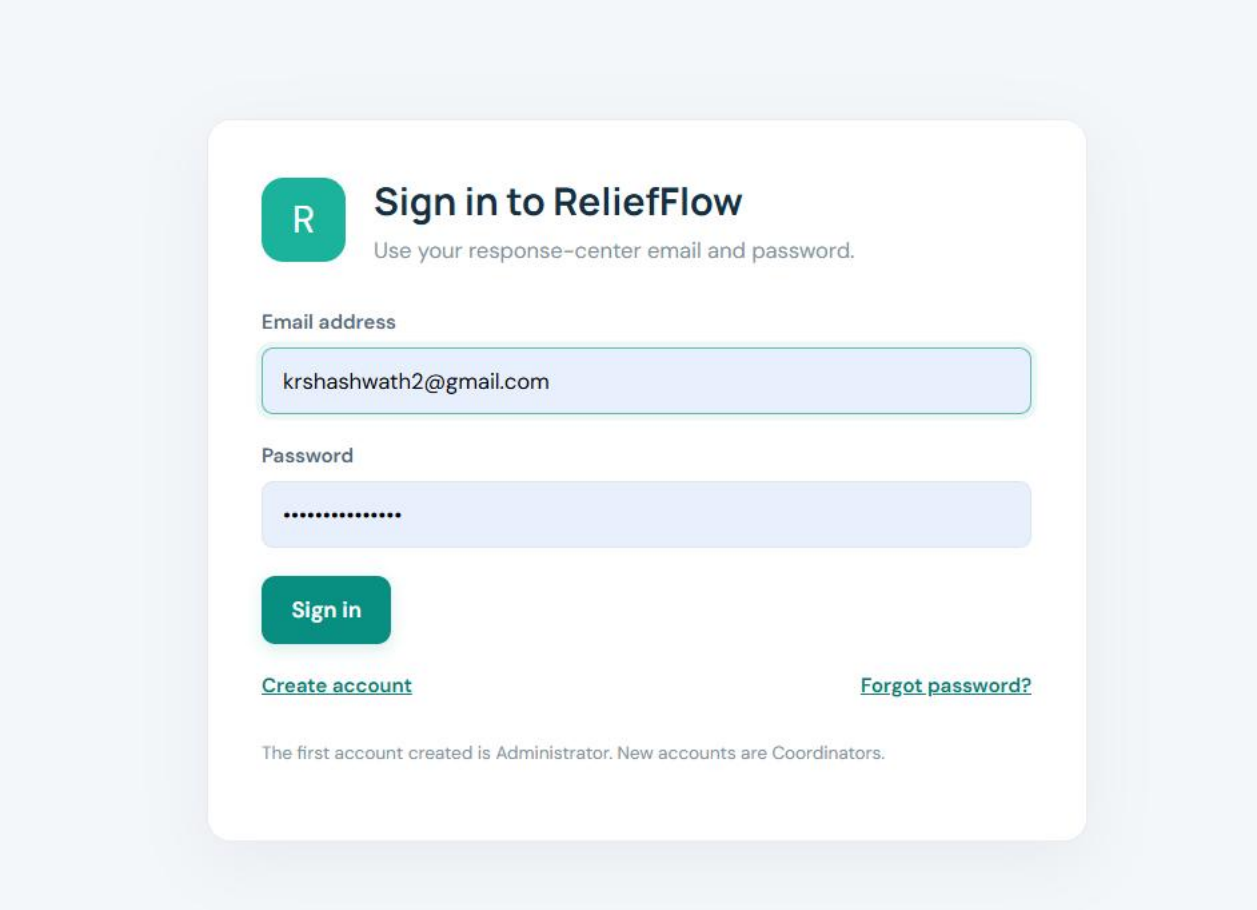


Figure 9.2: Create account page – a new user registers with full name, email and password.

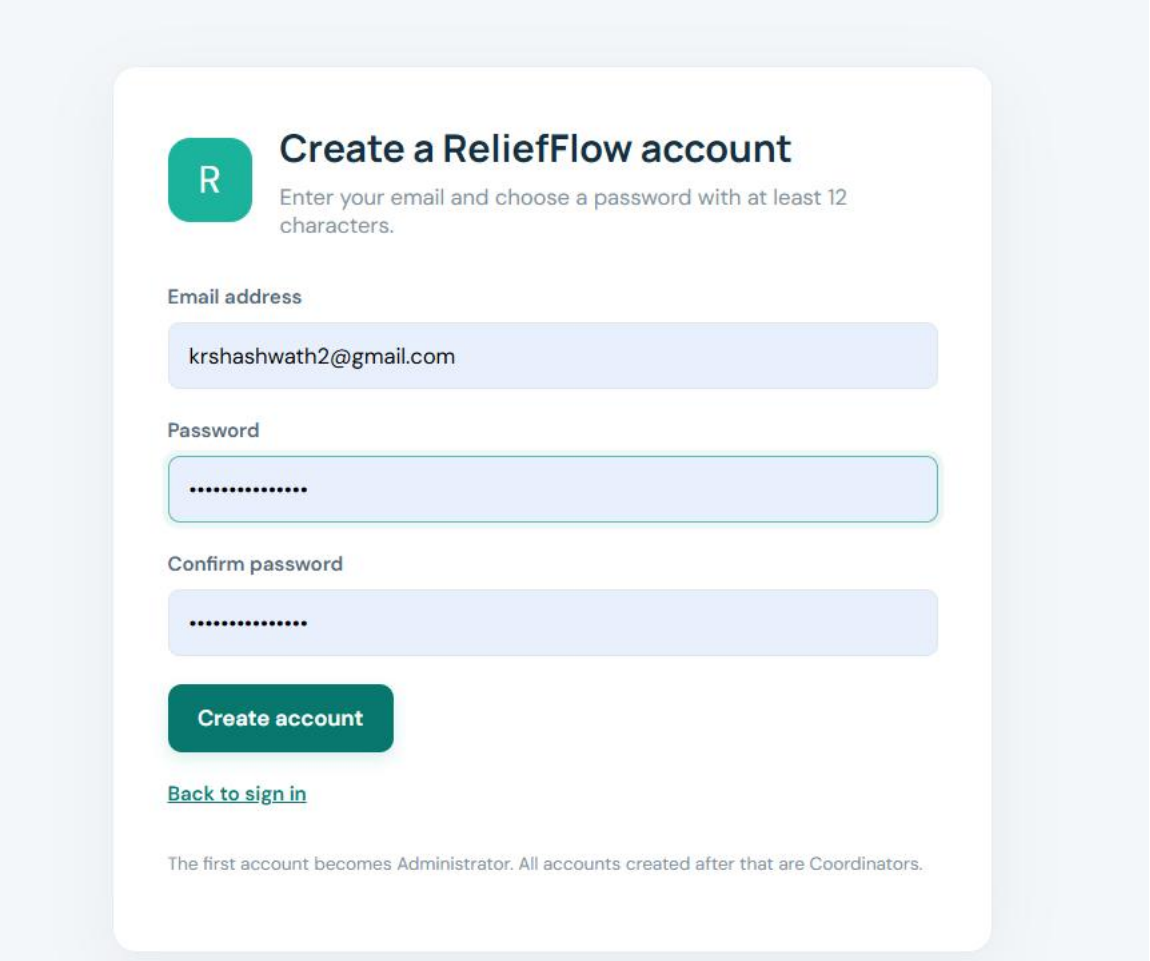
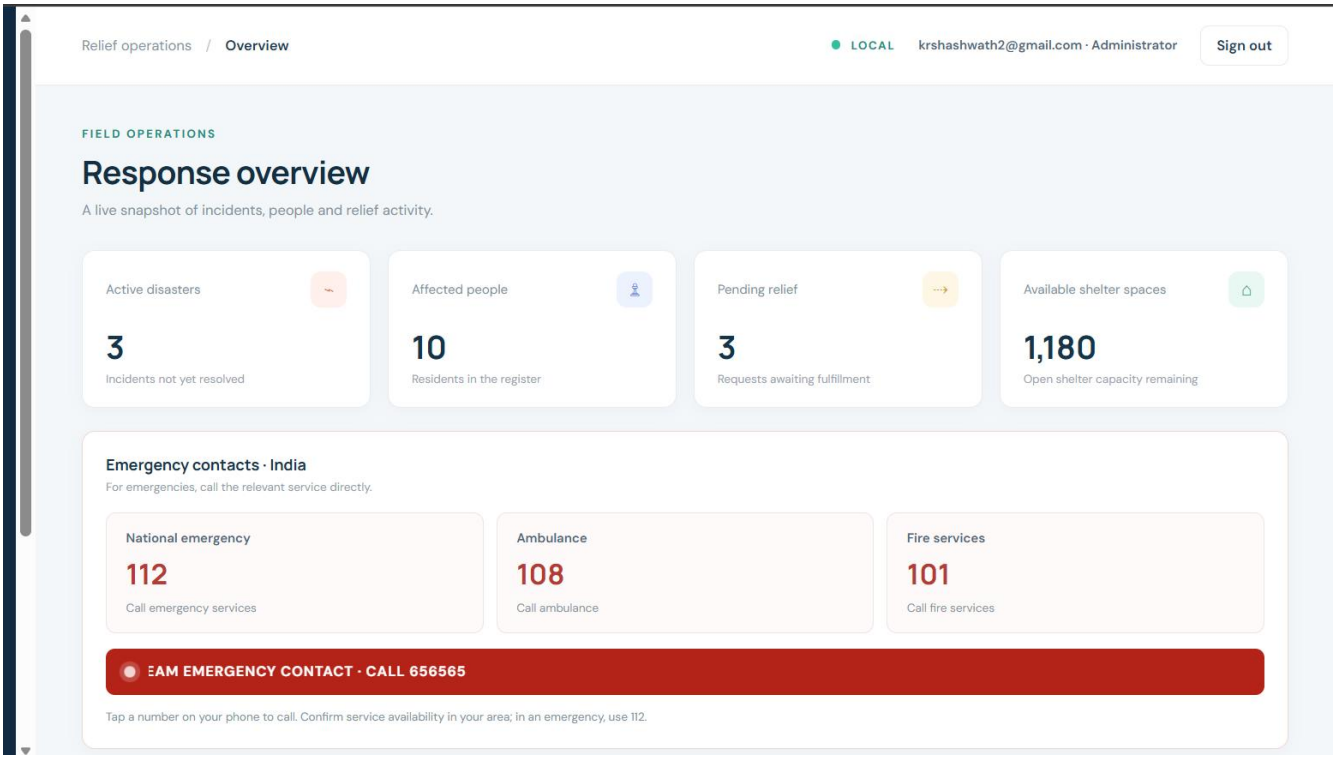
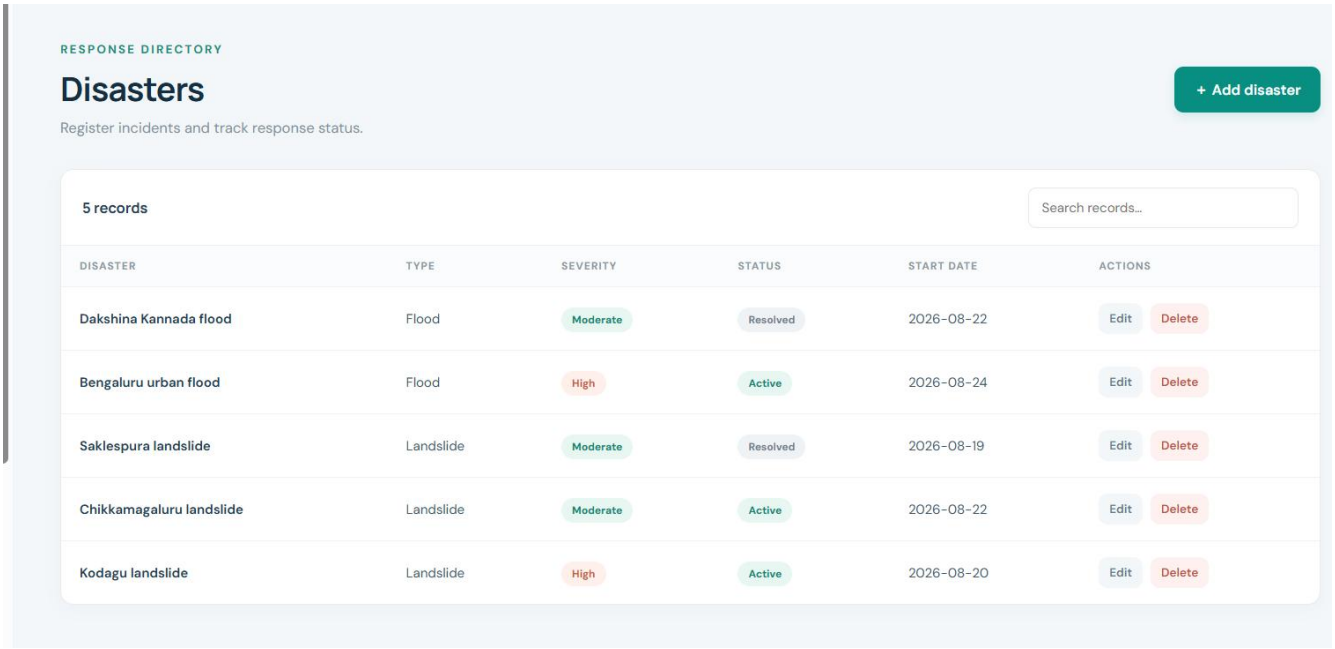


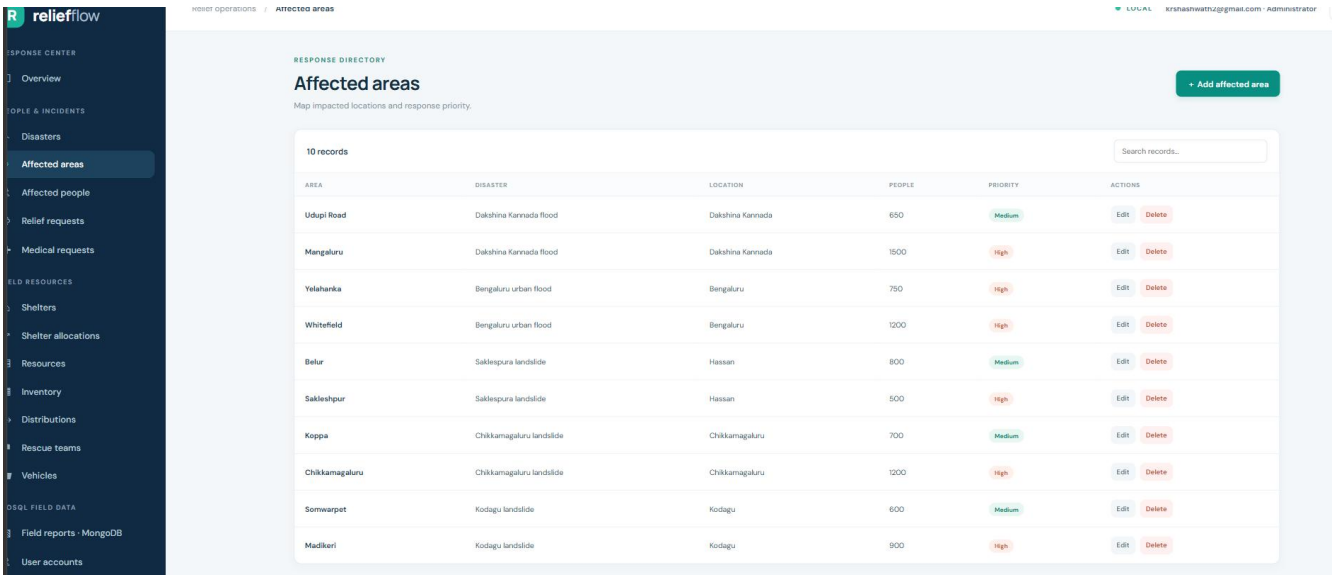
Figure 9.3: Home page



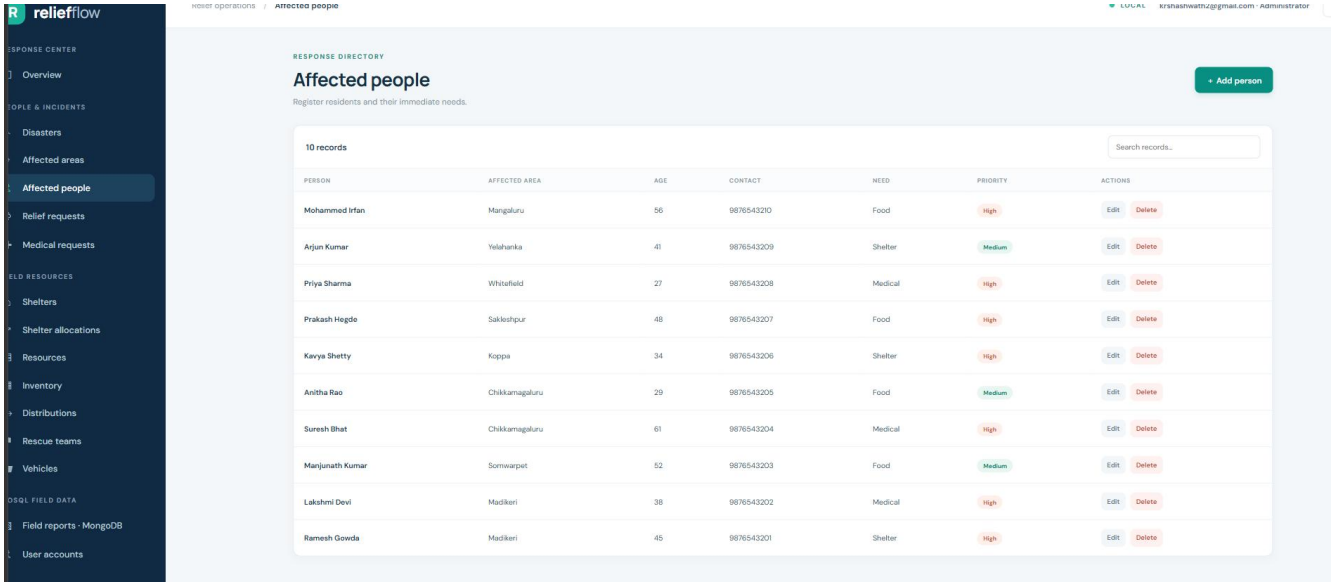
9 .4 Disaster - Basic details about each disaster, such as its name, type, location, and dates



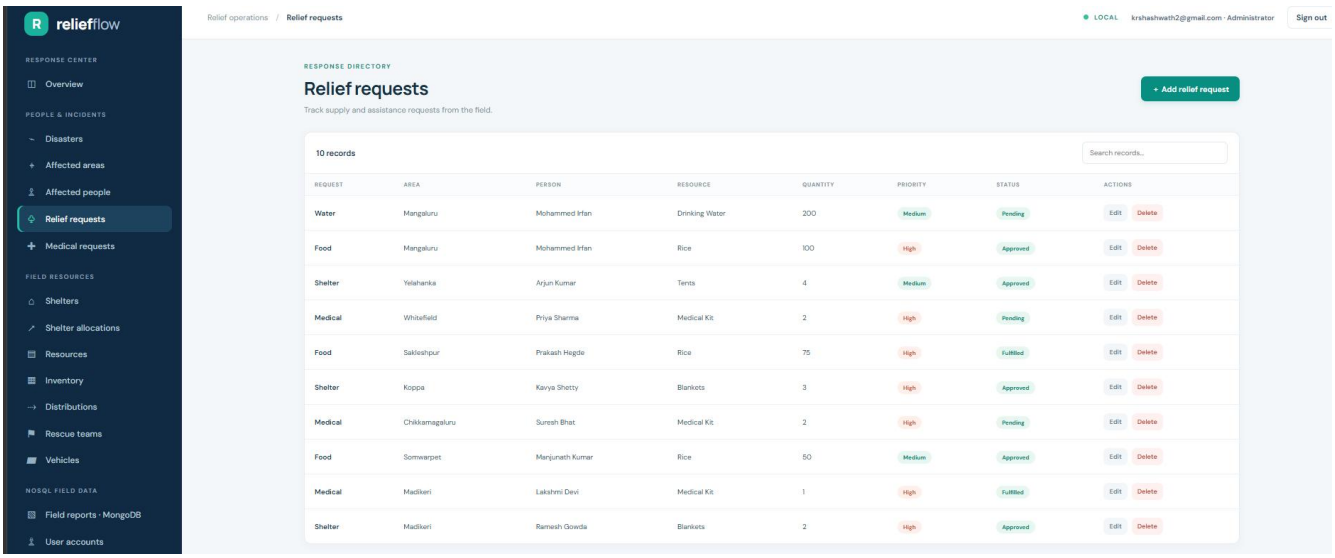
9 .5 Affected Area - Places affected by a disaster.



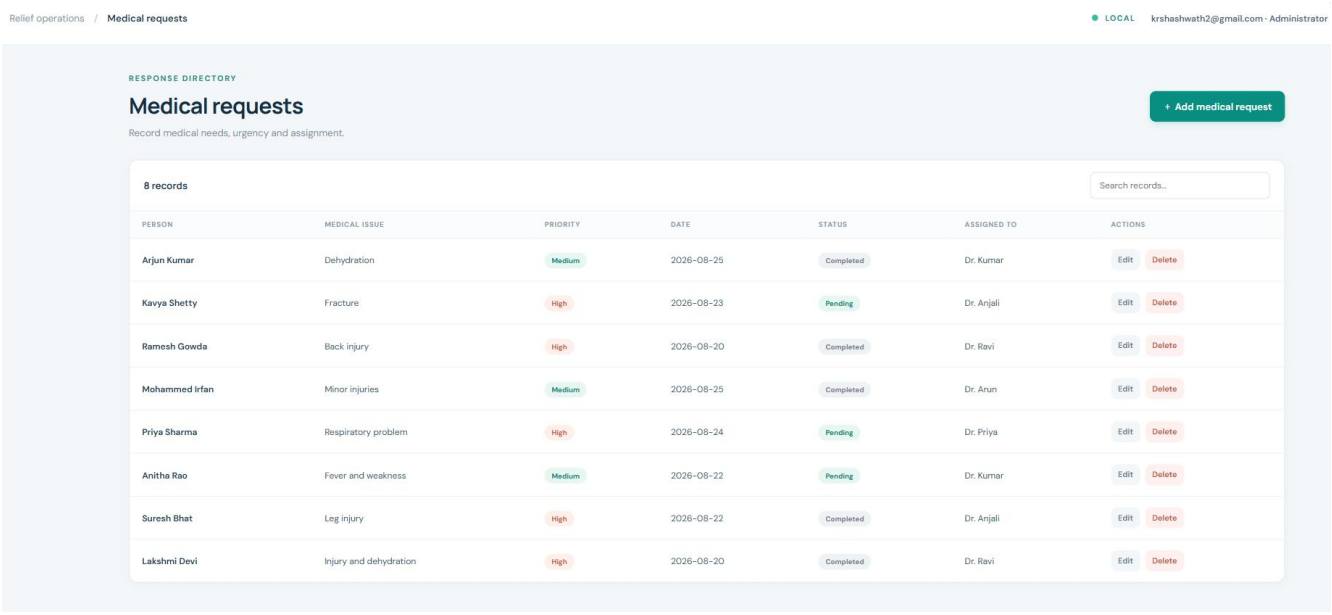
9 .6 Affected Person - Details of people affected by a disaster and their needs.



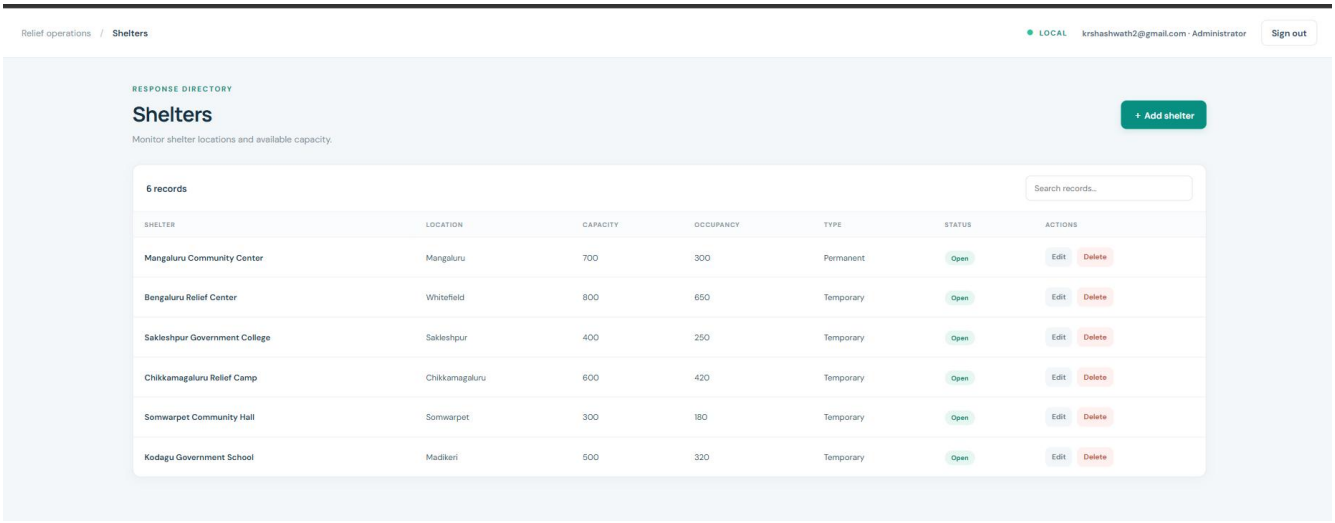
9 .7 Relief Request - Requests for food, water, or other relief supplies.



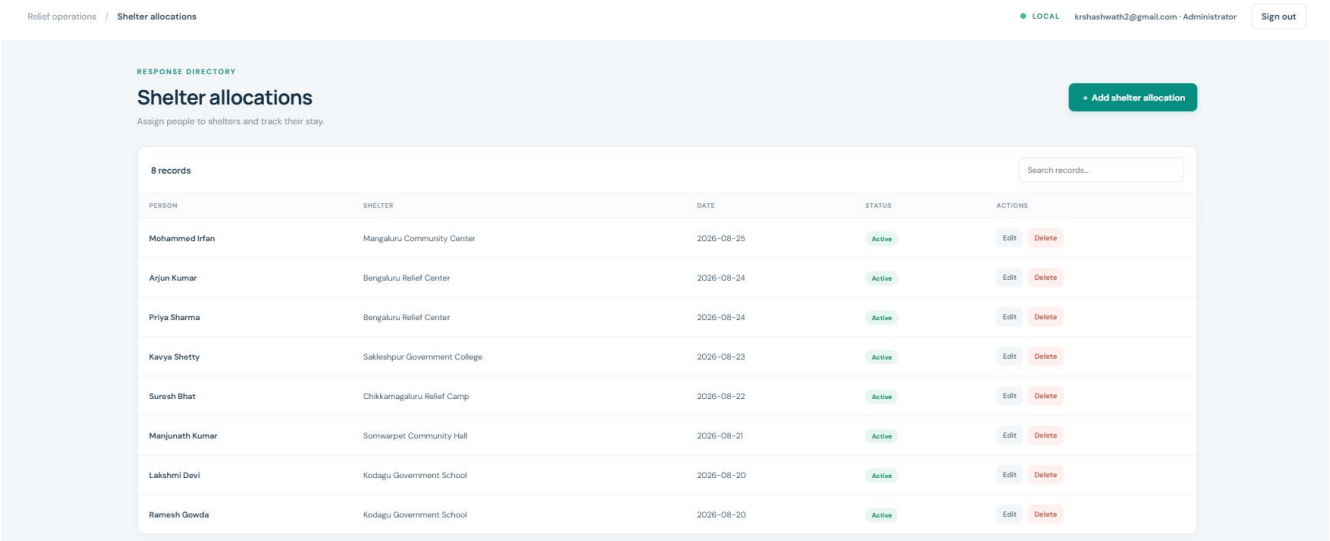
9.8 Medical Request - Medical help and treatment requests.



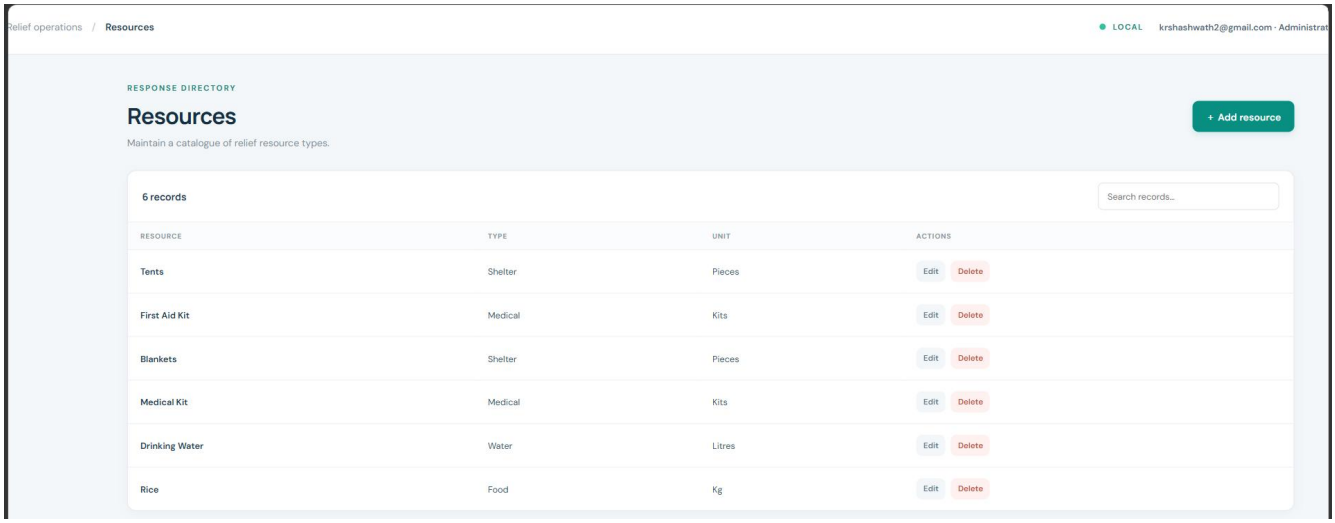
9.9 Shelter - Shelter locations and their capacity.



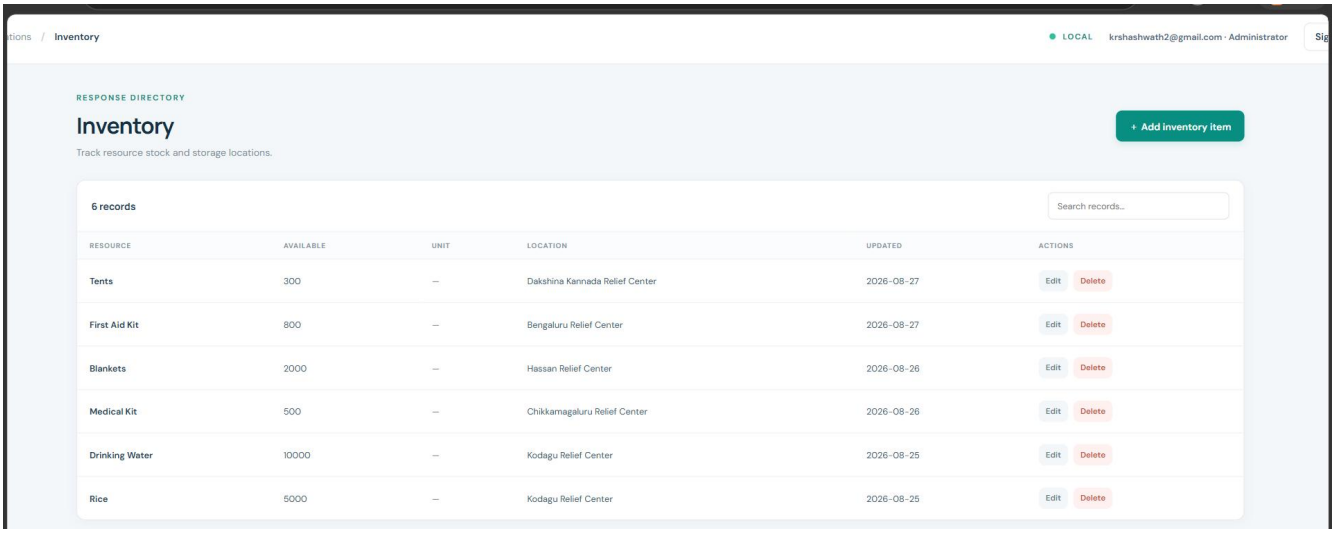
9 .10 Shelter Allocation - Records of people assigned to shelters.



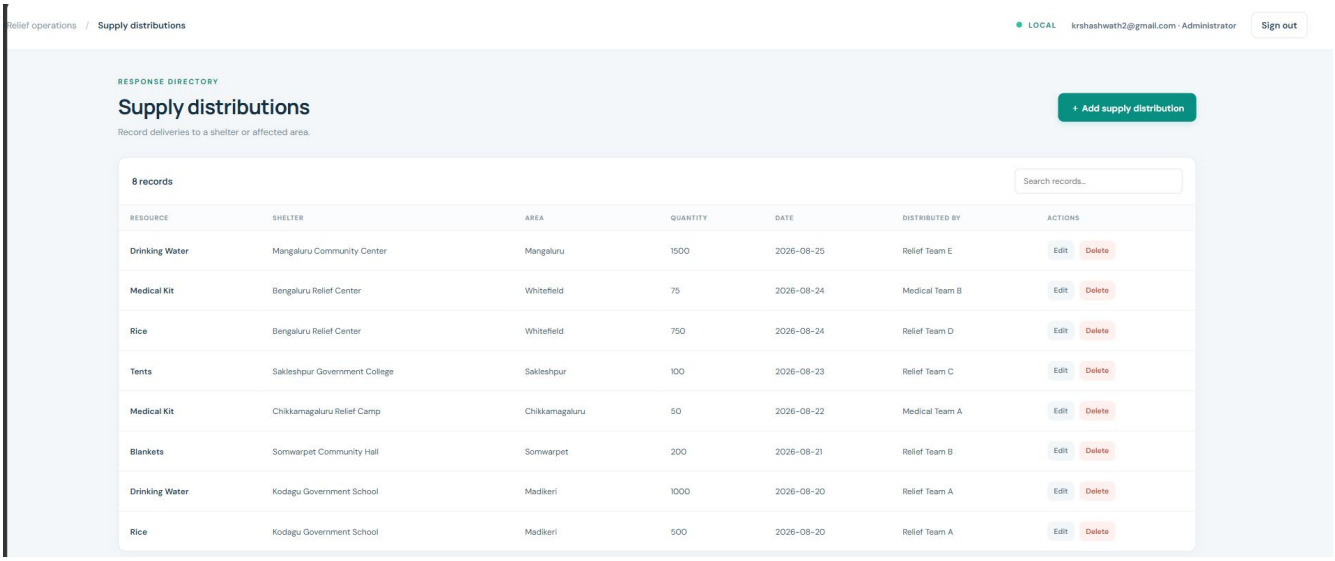
9 .11 Resource - Types of relief supplies available



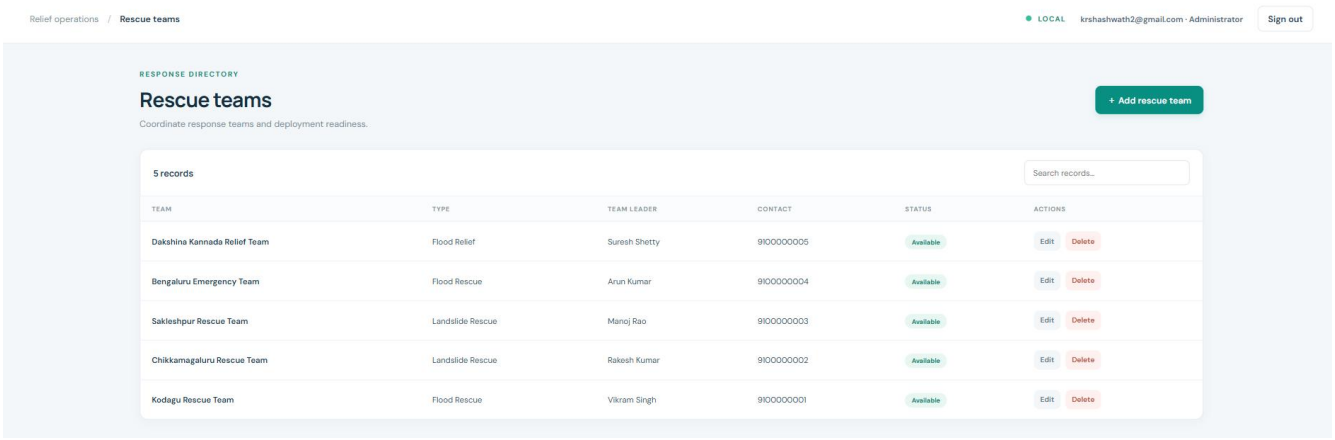
9. 12 Inventory - The quantity of each resource currently in stock.



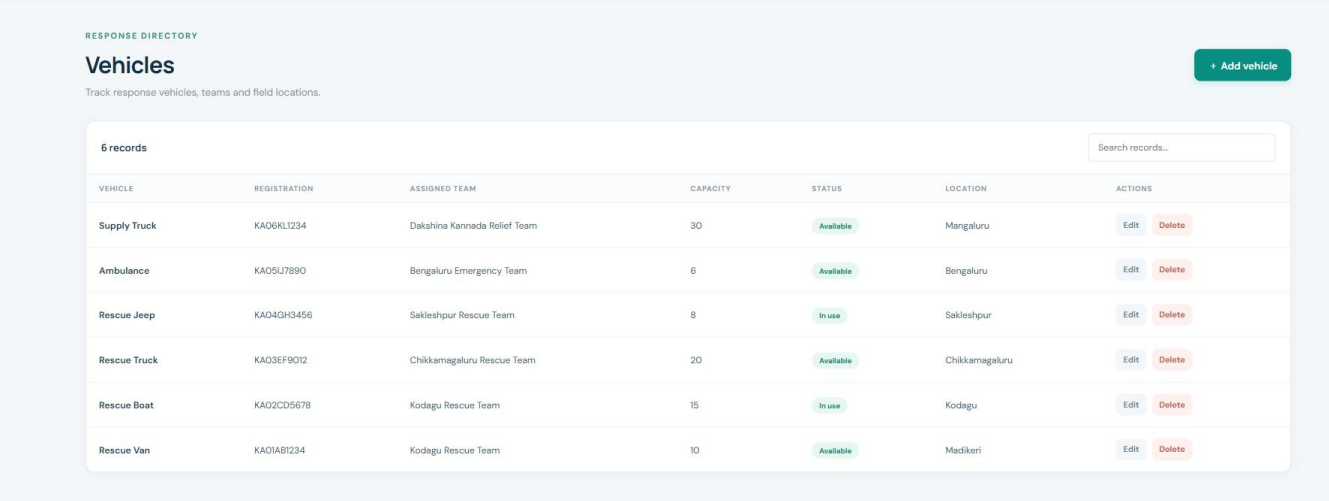
9 .13 Supply Distribution - Records of supplies sent to affected areas or shelters.



9.14 Rescue Team - Details about teams working on rescue operations.



9.15 Vehicle - Vehicles used by rescue and relief teams.



GitHub Link: https://github.com/krshashwath2-spec/DBMS-MINI-PROJECT

APPENDIX A: DEFINITIONS, ACRONYMS AND ABBREVIATIONS

Term	Meaning
DBMS	Database Management System
SQL	Structured Query Language
DDL	Data Definition Language (CREATE, ALTER, DROP)
DML	Data Manipulation Language (INSERT, UPDATE, DELETE)
ER	Entity-Relationship
PK	Primary Key
FK	Foreign Key
CHECK	Constraint that restricts the values allowed in a column
View	A stored query that behaves like a virtual table
Trigger	A program that runs automatically when a table is changed
SRN	Student Registration Number